

Constraint Architectures for Coherent Evaluability

Quotient Interfaces, Kernel Instantiation, and Structural Persistence

Amos Jay Maley

Abstract

We develop a constraint architecture for coherent evaluability in fixed-domain admissible evaluation systems. The framework is not presented as a permissive taxonomy of evaluative regimes, but as a structural forcing architecture: non-degenerate coherent evaluability imposes preservation constraints whose surviving kernel induces admissibility structure, transport covariance, quotient-interface formation, and structural persistence conditions.

The theory is built over admissibility-preserving evaluation systems equipped with witness extraction, identity-preserving transport, fail-closed enforcement, and irrecoverable standing failure. We prove that coherent evaluability forces quotient-interface structure, admissibility-preserving transport covariance, and kernel instantiation under fixed-domain continuity constraints. Auxiliary admissibility-bearing classifiers collapse to tensor-determined bookkeeping, no deeper same-domain standing-bearing layer survives, and no admissibility-preserving hidden evaluative remainder exists beneath the induced quotient interface.

Within the resulting architecture, coherent trajectories are invariant under admissibility-preserving transformation, closed under prefix extraction and legal composition, and fail-closed under standing loss. Symmetric self-location without invariant asymmetry is impossible, post-selection repair collapses, and admissibility-preserving operator totalization fails. The resulting quotient-interface structure is rigid up to admissibility-preserving interface equivalence.

The framework therefore derives coherent evaluability as a constrained structural regime rather than an arbitrary evaluative taxonomy. Alternatives incompatible with admissible continuity, witness coherence, and transport stability fail instantiation within the architecture itself.

Contents

1	Introduction	2
2	Fixed-Domain Admissible Evaluation Systems	4
2.1	States, presentations, and identity-preserving redescriptions	4
2.2	Layered equivalence hierarchy	7
2.3	Grounded transport	10
3	Inevitability of the Admissible Interface Conditions	11
4	AASC Full-Import Closure Layer	14
5	Projection as Quotient Descent	19
5.1	Admissible orbit equivalence	19
5.2	Transition descent	20
6	Persistence from Well-Founded Eligible Transport	22
6.1	Origins and derivative bearers	22
6.2	Rank without the shortest-chain error	23

7	Symmetric Branching and Equivariant Choice Failure	24
7.1	Branch points and local symmetry	24
7.2	Equivariant selectors	24
7.3	Worked toy model: symmetric representative choice	26
7.4	Contrasting near-failure example	27
8	Normal Form for Branch-Resolution Attempts	27
8.1	Resolution mechanisms	27
8.2	Four primary classes	27
9	Non-Duplication of Singular Identity Continuity	29
9.1	Identity-continuity claims	29
10	Anchor–Tensor–Skin Decomposition	30
10.1	Effect tests	30
11	Rigidity of the Reduced Interface	31
11.1	Projection layers and morphisms	31
12	Unified Structural Interface Theorem	34
13	Referee-Facing Audit	35
13.1	Dependency map	35
13.2	Standard objections and exact burdens	38
14	Relation to Existing Structural Frameworks	38
15	Discussion	39
A	Notation Crosswalk	40
B	Tightened Proof Seams	40
C	AASC Import Ledger for v8	42

1 Introduction

The paper studies a structural interface class rather than a biological or phenomenological object. A system is assumed to contain states, a witness-extraction basis determining fixed-domain evaluation signatures, identity-preserving changes of presentation, redescription-covariant licensed transition schemas, and a binary status marking which states may participate in the same fixed-domain evaluation regime. The question is what follows once that status is invariant under admissible presentation change, conserved by licensed transition, closed against external parameters, and grounded in a non-stipulative transport structure.

The intended reading is constraint-architectural. The paper is not offering a menu of possible evaluative systems and then selecting one attractive member. It asks what any non-degenerate coherent evaluability problem must instantiate if it is to remain determinate under witness extraction, neutral under presentation change, stable under continuation, internal to the represented domain, and non-stipulative in its continuity claims. Under that reading, the exclusions below

are not discretionary prohibitions. They are instantiation failures: a purported alternative that violates the kernel does not represent another admissible solution inside the same regime, but a collapse of the regime, a hidden parameter, a boundary failure, bookkeeping only, or an explicit change of problem.

The terminology is intentionally mainstream. The main devices are quotient maps, groupoid-like redescription actions, layered equivalence relations, coherent evaluation signatures, redescription-covariant transition relations, equivariant choice functions, well-founded predecessor relations, normal-form exhaustion, and admissible morphisms of reduced interfaces. The labels “projection”, “persistence”, “self-location”, and “identity” are used only as structural role labels. Identity-continuity in this paper means transport-preserving evaluative continuity; it carries no independent metaphysical or psychological commitment. The framework is intentionally minimal but closure-complete: additional structure is introduced only when required to preserve evaluative coherence under admissible transport. The kernel is therefore not an optional modeling layer. It is the common constraint residue left when non-degenerate coherent evaluability is required to remain fixed-domain, witness-determinate, presentation-neutral, internally supported, fail-closed, and transport-grounded.

The guiding claim is this:

If evaluation is represented by a coherent fixed-domain witness signature, redescription-neutral, internally closed, binary, fail-closed, independently witnessable, and transport-grounded, then the system factors through a reduced quotient interface; eligibility-positive persistence has non-derivative origins; symmetric singular representative selection is impossible without invariant asymmetry; attempted escapes reduce to a short priority-normalized normal form; singular continuity cannot be duplicated across symmetry-related presentation-distinct bearers without invariant asymmetry or regime violation; admissibility-relevant roles decompose into anchors, tensors, and skins; and the reduced interface is unique, locally and up to admissible conjugacy, once the invariant profile is fixed.

The final closure layer is an inevitability layer rather than another conditional strengthening. The paper first states the fixed-domain admissible evaluation structure explicitly, then proves that the same structure is forced by any non-degenerate fixed-domain evaluative continuity problem that demands determinate witness consequences, redescription-neutral presentation, schema-stable continuation, internal data dependence, and finite continuity witnesses. Thus the admissibility machinery is not an optional decoration placed on top of evaluation; it is the minimal closure structure required for evaluation to remain well-defined under those preservation demands.

The AASC import layer closes the boundary and operator side of the substrate. Boundary work is classificatory rather than operational: no selector, metric, ranking, probability, typicality rule, canonical representative, or discrete index has admissibility-bearing authority at the AMetric boundary unless it is already represented as a declared witness, anchor, or transported transition-label invariant. Standing and admissibility are not allowed to survive as competing same-domain gates. Same-scope operators are treated as certified envelopes: if they preserve the fixed invariant structure they are bookkeeping-equivalent, and if they totalize, repair, diagonalize, complete, or import hidden data they leave the regime. Thus the closure structure is not a patch around individual objections; it is the fixed-domain residue left after boundary, gate, operator, quotient, trajectory, and role loci are all exhausted.

Compared with the earlier proof-granularity version, the present version makes five hardening moves. First, admissible redescriptions are made explicitly groupoid-like, so that the quotient relation is a standard orbit equivalence. Second, the persistence proof no longer assumes that

arbitrary predecessors lie on shortest origin chains; instead, grounded persistence is expressed by well-founded strict transport, with ordinal rank available when desired. Third, branch selection is formulated as the standard obstruction to equivariant choice under a transitive symmetry. Fourth, escape exhaustion is stated as a priority-normalized normal-form theorem, avoiding overclaiming about mutually exclusive raw mechanisms. Fifth, the anchor–tensor–skin result is stated as an effect-based role decomposition on a fixed domain.

Scope and non-claims

The theorem does not claim that every formal system, every computer, or every physical organism satisfies the hypotheses. The framework classifies invariant continuity structures within admissible evaluation systems and does not assume any independent metaphysical theory of persons, consciousness, or ontology. It does not explain consciousness phenomenologically. It does not reduce psychology to syntax. It does not infer biological mentality from computation alone. The claim is not that all systems are covered, but that a system purporting to instantiate non-degenerate coherent evaluability within one fixed domain must satisfy the kernel constraints or cease to instantiate that regime. Once those constraints are in force, the structural roles associated with persistence, projection, self-location, and continuity are forced. The term “mind-like” is therefore shorthand for this role profile only.

2 Fixed-Domain Admissible Evaluation Systems

2.1 States, presentations, and identity-preserving redescrptions

Definition 2.1 (State space and alpha-identity). *Let S be a nonempty set of well-typed states. An equivalence relation $\sim$ on S is called identity-preserving when $X \sim Y$ means that X and Y differ only by a permitted change of presentation, such as capture-avoiding renaming of typed names, with no change in the admissibility-relevant structure.*

The restriction to a set avoids size issues in quotient formation. Nothing essential changes if one works in a universe large enough to contain the relevant classes.

Definition 2.2 (Admissible witness-extraction basis). *An admissible witness-extraction basis is a family*

$$W = \{w_i : S \multimap A_i\}_{i \in I}$$

of partial fixed-domain tests or extractors, with a distinguished undefined value $\perp_i$ adjoined to each codomain, satisfying the following requirements.

- (W1) **Internal support.** *The definedness and value of each $w_i(X)$ are determined by typed data represented in X and by fixed transition-label data, not by a meta-level parameter.*
- (W2) **Redescription covariance.** *For every admissible redescription R , each witness value is carried to the corresponding codomain presentation without changing its invariant content. In particular, equality of witness values up to codomain presentation is preserved by R .*
- (W3) **Finite extractability.** *Whenever a licensed mechanism consults a datum in order to decide eligibility, transport, or branch-relevant structure, that datum is recoverable by finitely many witnesses in W .*

(W4) **Separation of fixed-domain consequences.** *If two states differ in any admissible eligibility test, anchor test, tensor test, branch-invariant test, predecessor profile, or licensed fixed-domain consequence, then some witness in $\mathbf{W}$ distinguishes them.*

The basis may be infinite, but it is fixed for the domain under discussion and is not enlarged during a same-domain evaluation.

The basis is not assumed to encode all semantic, psychological, or metaphysical structure. It contains only operations needed to extract fixed-domain admissibility, eligibility, transport, and branch-relevant consequences.

Definition 2.3 (Witness minimality and consequence-complete subbasis). *For a witness basis $\mathbf{W}$, write*

$$X \equiv_{\mathbf{W}} Y \iff \forall w_i \in \mathbf{W}, w_i(X) \cong_i w_i(Y).$$

A subbasis $\mathbf{W}_0 \subseteq \mathbf{W}$ is consequence-complete when agreement on $\mathbf{W}_0$ implies agreement on every fixed-domain admissibility-relevant consequence separated by $\mathbf{W}$: eligibility, anchor tests, tensor tests, branch-invariant tests, predecessor profiles, and licensed quotient-transition consequences. The basis $\mathbf{W}$ is witness-minimal when no proper subbasis of $\mathbf{W}$ is consequence-complete. A witness is redundant exactly when deleting it leaves a consequence-complete subbasis.

Lemma 2.4 (Witness redundancy invariance). *Let $\mathbf{W}_0 \subseteq \mathbf{W}$ be consequence-complete. Replacing $\mathbf{W}$ by $\mathbf{W}_0$ leaves the admissibility-relevant quotient kernel, descended eligibility, quotient transition relation, predecessor relation, and anchor–tensor–skin role decomposition unchanged, up to harmless recoding of the signature codomain.*

Proof. Consequence-completeness says that whenever X and Y agree on $\mathbf{W}_0$, they agree on every fixed-domain consequence that $\mathbf{W}$ is permitted to separate. Hence deletion of redundant witnesses changes neither the equivalence relation generated by admissibility-relevant tests nor any relation defined from those tests. Descended eligibility, quotient transition, predecessor structure, and role classification are defined only from those consequences. Therefore the reduced interface is unchanged except for replacing the signature tuple by an isomorphic smaller tuple. $\square$

Definition 2.5 (Consequence-equivalent witness bases). *Let $\mathbf{W}$ and $\mathbf{W}'$ be two admissible witness-extraction bases on the same typed domain, with induced evaluation maps $\mathcal{E}_{\mathbf{W}} : S \rightarrow \Sigma_{\mathbf{W}}$ and $\mathcal{E}_{\mathbf{W}'} : S \rightarrow \Sigma_{\mathbf{W}'}$. They are consequence-equivalent, written*

$$\mathbf{W} \equiv_{\text{Con}} \mathbf{W}',$$

when they generate the same fixed-domain admissibility-relevant consequence kernel: for every $X, Y \in S$, agreement of X and Y on all witnesses in $\mathbf{W}$ holds exactly when agreement on all witnesses in $\mathbf{W}'$ holds, and this common agreement is exactly agreement on eligibility, anchor tests, tensor tests, branch-invariant tests, predecessor profiles, licensed quotient-transition consequences, and every fixed-domain consequence available to certified mechanisms. Consequence-equivalence permits harmless recoding of codomains but not insertion or deletion of consequence-relevant information.

Theorem 2.6 (Witness-basis equivalence). *Let $\mathcal{M}_{\mathbf{W}}$ and $\mathcal{M}_{\mathbf{W}'}$ be two grounded fixed-domain admissible evaluation systems on the same typed domain, differing only by replacing $\mathbf{W}$ with $\mathbf{W}'$ and by the induced recoding of $\mathcal{E}$, and suppose $\mathbf{W} \equiv_{\text{Con}} \mathbf{W}'$. Assume also that every admissibility certificate valid relative to one basis is valid relative to the other after the codomain recoding induced*

by consequence-equivalence. Then the two systems have the same admissibility-relevant quotient kernel, the same descended eligibility, the same quotient transition relation, the same quotient predecessor relation, the same anchor–tensor–skin role classification, and reduced projection interfaces equivalent up to admissible conjugacy.

Proof. Consequence-equivalence says that the two bases induce the same kernel on all fixed-domain consequence tests. Hence they agree on which state pairs may be identified without losing eligibility, anchor, tensor, branch-invariant, predecessor, or licensed transition information. The additional certificate assumption ensures that the admissible redescription and transition schemas are not changed except by harmless codomain recoding. Therefore the orbit quotient and the minimal evaluation-equivalence kernel coincide for the two systems. Descended eligibility, quotient transition, quotient predecessor, and the role tests are all defined from those invariant consequences, so they coincide as well. The induced signature images may be notationally different, but consequence-equivalence supplies a bijective recoding between them. Taking that recoding as the signature component and the identity on the common quotient classes gives an admissible interface conjugacy. $\square$

Remark 2.7 (Canonicity of witness choice). *The witness basis is therefore not canonical as a list of extractors, but it is canonical up to consequence-equivalence. Different bases that generate the same invariant evaluative consequences yield the same reduced interface; bases that do not generate the same consequences define different regimes rather than different presentations of one regime.*

Remark 2.8 (Semantic loading blocked). *If a proposed witness carries extra semantic decoration that does not affect fixed-domain consequence extraction, Lemma 2.4 removes it without changing the reduced interface. If it does affect those consequences, it is not decoration; it changes the evaluation regime and must satisfy the witness and admissibility criteria explicitly.*

Definition 2.9 (Evaluation signature, functional, and coherence). *Let*

$$\Sigma \subseteq \prod_{i \in I} (A_i \cup \{\perp_i\}) / \cong_i$$

be the image of the witness basis modulo harmless codomain presentation equivalence. The evaluation functional is

$$\mathcal{E} : S \rightarrow \Sigma, \quad \mathcal{E}(X) = ([w_i(X)]_{\cong_i})_{i \in I}.$$

It satisfies the evaluation coherence condition

$$\mathcal{E}(X) = \mathcal{E}(Y) \iff \forall i \in I, w_i(X) \cong_i w_i(Y).$$

Equivalently, equality of evaluation signatures means equality of every admissible fixed-domain witness consequence. Thus Σ is not an arbitrary bookkeeping codomain chosen after the fact; it is the quotient image of a fixed witness-extraction basis.

Lemma 2.10 (Evaluation coherence). *If $\mathcal{E}(X) = \mathcal{E}(Y)$, then no licensed mechanism whose data access is restricted to the witness basis can distinguish X from Y by eligibility, anchor, tensor, branch-invariant, predecessor, or fixed-domain consequence data. If such a mechanism can distinguish them, then $\mathcal{E}(X) \neq \mathcal{E}(Y)$.*

Proof. The forward direction follows from the coherence condition: every datum available to a licensed mechanism is represented by finitely many witnesses, and all such witness values agree. The converse follows from witness separation: any admissible fixed-domain distinction is detected by some w_i , hence appears in the signature tuple. Therefore the evaluation functional is constrained by witness consequences rather than by an unconstrained choice of codomain. $\square$

Remark 2.11 (Evaluation is not left intuitive). *The word “evaluation” is used below only through the witness basis $\mathcal{W}$, the induced signature map $\mathcal{E}$, Elig, licensed transition schemas, and the induced quotient interface. Thus the framework does not rely on an informal background notion of evaluation.*

Definition 2.12 (Admissible redescription groupoid). *A class $\mathcal{R}$ of total maps $R : S \rightarrow S$ is an admissible redescription groupoid modulo $\sim$ when:*

- (R1) $\text{id}_S \in \mathcal{R}$;
- (R2) if $R_1, R_2 \in \mathcal{R}$, then $R_2 \circ R_1 \in \mathcal{R}$;
- (R3) for every $R \in \mathcal{R}$ there is $R^{-1} \in \mathcal{R}$ such that $R^{-1}R(X) \sim X$ and $RR^{-1}(X) \sim X$ for all $X \in S$;
- (R4) each $R \in \mathcal{R}$ preserves typing, the fixed-domain signature, the evaluation functional $\mathcal{E}$, the witness basis $\mathcal{W}$ up to codomain presentation, and the admissibility-relevant envelope;
- (R5) $X \sim Y$ implies $R(X) \sim R(Y)$ for every $R \in \mathcal{R}$.

Elements of $\mathcal{R}$ are called admissible redescrptions or identity-preserving presentation transformations.

2.2 Layered equivalence hierarchy

Definition 2.13 (Four equivalence layers). *The paper distinguishes four equivalence notions.*

- (E1) **Alpha or immediate presentation identity** $\equiv_\alpha$ *is the base identity-preserving relation $\sim$.*
- (E2) **Admissible presentation orbit equivalence** $\equiv_{\text{orb}}$ *holds when two states are related by an admissible redescription, modulo $\sim$:*

$$X \equiv_{\text{orb}} Y \iff \exists R \in \mathcal{R} R(X) \sim Y.$$

This is the quotient relation denoted $\approx$ below.

- (E3) **Transport equivalence** $\equiv_{\text{tr}}$ *holds when two states have the same eligibility value and the same predecessor/continuation profile under all licensed transitions, after passing to admissible presentation orbits.*
- (E4) **Evaluation equivalence** $\equiv_{\text{ev}}$ *holds when no fixed-domain evaluation signature, anchor test, tensor test, eligibility verdict, predecessor relation, or induced transition history distinguishes the states.*

The inclusions used in the paper are methodological rather than ontological: $\sim$ records immediate presentation identity, $\approx$ removes admissible presentation variation, and minimality later identifies the quotient kernel with the evaluation-equivalence kernel.

Remark 2.14 (No semantic slippage). *The quotient used for projection is not an undefined mixture of syntactic, semantic, observational, and metaphysical sameness. It is the admissible presentation-orbit relation $\equiv_{\text{orb}}$. Evaluation equivalence appears only in the later minimality condition, where it is explicitly defined as the common kernel of the invariant tests available inside the fixed domain.*

Definition 2.15 (Transition-schema covariance action). *A class $\mathcal{T}$ of maps $T : S \rightarrow S$ is a class of licensed transition schemas when every $T \in \mathcal{T}$ is well-typed and acts within the same fixed domain, and the redescription groupoid acts on $\mathcal{T}$ by a covariance action*

$$\# : \mathcal{R} \longrightarrow \text{Bij}(\mathcal{T}), \quad R \mapsto R_{\#},$$

satisfying, modulo pointwise $\sim$ on transition outputs,

$$(\text{id}_S)_{\#} = \text{id}_{\mathcal{T}}, \quad (R_2 \circ R_1)_{\#} = R_{2\#} \circ R_{1\#}, \quad (R^{-1})_{\#} = (R_{\#})^{-1}.$$

The covariance law is

$$R(T(X)) \sim (R_{\#}T)(R(X))$$

for every $R \in \mathcal{R}$, $T \in \mathcal{T}$, and $X \in S$. A schema may represent an update, continuation, evaluation step, or transport step. Moves that change the domain, silently import new parameters, repair a violation, or lack a witness certificate are not members of $\mathcal{T}$. A transition schema T is redescription-stable when $R_{\#}T = T$ for every $R \in \mathcal{R}$.

Lemma 2.16 (Covariance action coherence). *The $R_{\#}$ operation preserves identity, composition, and inverse redescription transport on transition schemas. Hence branch labels and transition labels are transformed as part of the same fixed-domain structure rather than being held as invisible external constants.*

Proof. The identities are part of the covariance-action specification. They ensure that applying no redescription leaves a transition schema unchanged, applying two redescriptions transports the schema in the corresponding composed order, and undoing a redescription also undoes its transition-label action. Therefore transition labels are not meta-level parameters; they are acted on by the same groupoid structure that acts on state presentations. $\square$

Definition 2.17 (Eligibility). *An eligibility predicate is a map*

$$\text{Elig} : S \rightarrow \{0, 1\}.$$

States with $\text{Elig}(X) = 1$ are eligible or standing-positive; states with $\text{Elig}(X) = 0$ are ineligible. Write

$$S^+ = \{X \in S : \text{Elig}(X) = 1\}, \quad S^- = \{X \in S : \text{Elig}(X) = 0\}.$$

Definition 2.18 (Fixed-domain admissible evaluation system). *A fixed-domain admissible evaluation system is a tuple*

$$\mathcal{M} = (S, \sim, \mathcal{E}, \mathcal{R}, \mathcal{T}, \text{Elig})$$

satisfying the following conditions:

(F1) **Witness-coherent evaluation.** *The functional $\mathcal{E}$ is induced by a fixed witness-extraction basis W satisfying evaluation coherence. Licensed mechanisms may consult only data represented by W , by $\mathcal{E}$, or by fixed transition-label data transformed under $R_{\#}$.*

(F2) **Redescription neutrality.** *For all $R \in \mathcal{R}$ and all $X \in S$,*

$$\text{Elig}(R(X)) = \text{Elig}(X).$$

Also, if $X \sim Y$, then $\text{Elig}(X) = \text{Elig}(Y)$.

(F3) **Transition covariance.** For all $R \in \mathcal{R}$, $T \in \mathcal{T}$, and $X \in S$,

$$R(T(X)) \sim (R_{\#}T)(R(X)).$$

Redescription-stable individual transitions are the special case $R_{\#}T = T$.

(F4) **Binary conservation and fail-closedness.** For all $T \in \mathcal{T}$ and all $X \in S$,

$$\text{Elig}(T(X)) = \text{Elig}(X).$$

Equivalently, licensed same-domain transition preserves positive eligibility and cannot internally promote an ineligible state. Attempted moves that would promote, repair, or silently reset eligibility are treated as outside the licensed transition class.

(F5) **Internal data closure.** If a mechanism computes $T(X)$, or computes a branch choice from X , using a datum δ , then δ is represented inside the state or inside the fixed transition signature and is acted on by admissible redescriptions. There are no evaluation-relevant meta-level parameters invisible to $\mathcal{R}$.

(F6) **Non-degeneracy.** $S^+ \neq \emptyset$ and $S^- \neq \emptyset$.

Definition 2.19 (Nontrivial admissibility criteria). A map is not admissible merely because it is a map on S . Admissibility is constrained by the following tests:

(A1) type and envelope preservation;

(A2) witness preservation or covariance relative to $\mathcal{W}$;

(A3) evaluation-signature preservation for redescriptions, $\mathcal{E}(RX) = \mathcal{E}(X)$;

(A4) eligibility preservation, $\text{Elig}(FX) = \text{Elig}(X)$, for same-domain licensed maps;

(A5) compatibility with transition covariance under the $R_{\#}$ action;

(A6) internality of all data used by the mechanism, meaning no branch-relevant or eligibility-relevant datum is invisible to $\mathcal{R}$.

These criteria are part of the mathematical specification, not consequences inferred after the fact.

Definition 2.20 (Independent admissibility witness certificate). A candidate redescription or transition schema is admitted only when there is a fixed-domain certificate, external to the theorem conclusions, verifying the relevant type, envelope, witness, eligibility, covariance, and internality clauses. For redescriptions this certificate shows invariant preservation of $\mathcal{W}$, $\mathcal{E}$, Elig , and the envelope. For transition schemas it shows that every consulted datum is witness-extractable and that redescription covariance holds with a specified action $R_{\#}$ on transition labels. The certificate is not allowed to appeal to the desired conclusions of quotient descent, non-duplication, or rigidity.

Lemma 2.21 (Admissibility nontriviality). In a non-degenerate fixed-domain admissible evaluation system, not every well-typed self-map of S is admissible. In particular, any map that changes eligibility, changes witness consequences, changes the evaluation signature in a redescription, injects an external branch-relevant parameter, lacks an independent witness certificate, or fails transition covariance is excluded from $\mathcal{R}$ or $\mathcal{T}$.

Proof. By non-degeneracy choose $X^+ \in S^+$ and $X^- \in S^-$. A self-map F with $F(X^+) = X^-$ changes eligibility and therefore violates the eligibility-preservation test. A self-map G that changes some witness consequence changes $\mathcal{E}$ by Lemma 2.10, so it cannot be an admissible redescription. A mechanism that depends on a datum not represented by the witness basis and not acted on by admissible redescription violates internality. A transition schema for which no $R_\#$ action makes $R(TX) \sim (R_\#T)(RX)$ violates covariance. Hence admissibility excludes concrete pathological maps and is a constrained witness-certified structure rather than a tautological closure of all transformations. $\square$

Theorem 2.22 (No universal admissibility). *In any non-degenerate fixed-domain admissible evaluation system, the admissible redescription and transition classes cannot contain all well-typed self-maps of S .*

Proof. By non-degeneracy choose $X^+ \in S^+$ and $X^- \in S^-$. A well-typed self-map sending X^+ to X^- exists whenever the state type admits such a function, and any analogous partial construction can be represented as a candidate mechanism on the two-state subprofile. If such a map were an admissible redescription, it would violate redescription neutrality for Elig. If it were a licensed same-domain transition, it would violate binary conservation. Similarly, any self-map that changes a witness consequence violates evaluation coherence for redescriptions, and any map that uses a datum not transformed by $\mathcal{R}$ violates internal data closure. Therefore universal admissibility is incompatible with non-degeneracy, witness coherence, and fail-closed conservation. $\square$

Lemma 2.23 (Witness criterion blocks arbitrary signatures). *The evaluation signature space Σ cannot be freely changed to force a preferred quotient without changing the fixed-domain witness basis. Any such change either preserves all witness fibers and is a harmless recoding, or changes at least one witness consequence and therefore changes the regime.*

Proof. By definition, Σ is the image of the witness tuple modulo codomain presentation. A recoding that preserves all witness fibers induces an isomorphic signature space and leaves $\mathcal{E}$ unchanged up to notation. A recoding that merges or separates witness fibers changes whether some w_i distinguishes two states. By evaluation coherence, that changes the fixed-domain consequence structure and therefore describes a different evaluation regime rather than a free presentation choice. $\square$

Remark 2.24 (Why conservation is equality). *The earlier formulation separated positive preservation and absorption of failure. Because eligibility is binary, the two clauses together say exactly that licensed same-domain transitions preserve eligibility. This does not mean that failure is impossible; it means that a failure-producing or repair-producing operation is not a licensed same-domain transition in the same regime. It must be modeled as termination, violation, or explicit reset.*

2.3 Grounded transport

Eligibility would be mathematically cheap if it were an arbitrary label. The persistence results require a stricter condition: positive eligibility must be grounded in a well-founded transport/dependence structure rather than stipulated independently of admissible continuation.

Definition 2.25 (Strict eligible predecessor). *A binary relation $\prec$ on S^+ is a strict eligible predecessor relation when $X \prec Y$ means that X is an immediate admissible predecessor on which Y depends as the same fixed-domain eligible continuation. It is required that:*

(P1) $X \prec Y$ implies $X, Y \in S^+$;

- (P2) $X \prec Y$ is generated only by nontrivial licensed continuation or transport, never by the identity transition;
- (P3) $X \prec Y$ implies $X \not\approx Y$ whenever $\approx$ is the admissible quotient relation defined below, unless the transition label itself records a nontrivial continuation step; and
- (P4) $\prec$ is invariant under admissible redescription: if $X \prec Y$ and $R \in \mathcal{R}$, then $R(X) \prec R(Y)$ up to $\sim$.

Definition 2.26 (Grounded conservation). *A fixed-domain admissible evaluation system is grounded when it is equipped with a strict eligible predecessor relation $\prec$ that is well-founded: every nonempty subset $A \subseteq S^+$ has a $\prec$ -minimal element. This is the formal replacement for the informal requirement that positive eligibility not be an ungrounded or arbitrary primitive label.*

Remark 2.27 (Non-stipulativity made precise). *The phrase “non-stipulative standing” can be read in many ways. In this paper it is not left as a rhetorical condition. It is represented by the explicit mathematical requirement that eligible transport be well-founded. Under that requirement, every standing-positive dependency chain has a non-derivative base; without it, infinite regress or arbitrary labeling is possible.*

3 Inevitability of the Admissible Interface Conditions

The preceding section stated the formal structure used in the rest of the paper. This section records the AASC-style closure point in mainstream terms: the structure is not merely assumed for convenience. It is forced by the preservation requirements implicit in any non-degenerate fixed-domain evaluative continuity problem.

Definition 3.1 (Constraint architecture). *A constraint architecture is a formal regime in which the target class is not specified by listing admissible alternatives, but by propagating preservation requirements and retaining only structures that survive them. In the present paper the target class is non-degenerate coherent evaluability in a fixed domain. A purported alternative is not rejected because it is unfamiliar or unwanted; it is rejected inside the regime only when it fails witness determinacy, presentation neutrality, schema-stable continuation, internal support, fail-closed standing, grounded continuity, or one of their AASC closure consequences.*

Definition 3.2 (Kernel instantiation). *A system instantiates the kernel when it realizes the minimal package required by the present constraint architecture: witness-coherent evaluation, identity-preserving redescription, redescription-covariant continuation, standing/admissibility identity, fail-closed eligibility, internal data closure, grounded transport, boundary non-selection, certified operator envelopes, no hidden remainder, and anchor-tensor-skin role exhaustion. Kernel instantiation is not an additional conclusion placed on top of the interface theorem; it is the condition under which non-degenerate coherent evaluability is well-posed inside the fixed domain.*

Definition 3.3 (Fixed-domain evaluative continuity problem). *A fixed-domain evaluative continuity problem consists of a typed state domain S , a nontrivial binary participation status $P : S \rightarrow \{0, 1\}$, a class of presentation changes, a class of candidate continuation schemas, and a family of finite witness extractions sufficient to determine every same-domain consequence the problem treats as evaluatively relevant. The problem is non-degenerate when both $P^{-1}(1)$ and $P^{-1}(0)$ are nonempty. It is determinate when two presentations assigned the same witness consequences cannot be distinguished by any licensed same-domain evaluative procedure.*

Definition 3.4 (Continuity adequacy demands). *A fixed-domain evaluative continuity problem satisfies continuity adequacy when it imposes the following five demands.*

- (N1) **Witness determinacy.** *Every evaluatively relevant same-domain consequence is finitely extractable from the represented state or from represented transition-label data.*
- (N2) **Presentation neutrality.** *Mere change of admissible presentation cannot alter participation status or witness consequences.*
- (N3) **Continuation stability.** *Licensed continuation is compatible with presentation change: presenting first and continuing later must agree, up to presentation, with continuing first and presenting later after the continuation label is transported.*
- (N4) **No hidden parameters.** *Branch-relevant, eligibility-relevant, or continuation-relevant data may not remain invisible to presentation transformations.*
- (N5) **Finite continuity grounding.** *A positive continuity claim must admit a finite invariant witness profile or terminate in a non-derivative positive source; infinite regress without extractable grounding is not a determinate same-domain continuity claim.*

Lemma 3.5 (Witness determinacy forces evaluation coherence). *Any determinate fixed-domain evaluative continuity problem satisfying witness determinacy admits an evaluation functional $\mathcal{E} : S \rightarrow \Sigma$ whose equality relation is exactly equality of all fixed-domain witness consequences.*

Proof. Form the tuple of all witness consequences required by witness determinacy and quotient each witness codomain by harmless presentation equivalence. This tuple is $\mathcal{E}(X)$. If two states have equal tuples, determinacy says no licensed same-domain evaluative procedure distinguishes them. If a licensed procedure distinguishes them, witness determinacy supplies a finite witness that differs, so the tuples are unequal. Hence evaluation coherence is forced by determinacy rather than chosen freely. $\square$

Lemma 3.6 (Presentation neutrality forces an admissible redescription groupoid). *If presentation changes are to be neutral with respect to fixed-domain evaluative consequences, then the admissible presentation changes must be restricted to identity-preserving transformations closed under identity, composition, and inverse up to presentation equivalence.*

Proof. Neutral presentation change must include doing nothing, must remain neutral after two neutral changes are performed successively, and must not create irreversible evaluative information merely by changing notation. Therefore identity and composition are forced. If a presentation change is neutral, returning to the previous presentation must also be neutral up to the same presentation equivalence; otherwise the purported presentation change would have changed evaluative content. Thus inverse closure up to $\sim$ is forced. Preservation of witnesses, $\mathcal{E}$, Elig , and the envelope is exactly the formal content of neutrality. $\square$

Lemma 3.7 (Continuation stability forces transition-schema covariance). *Any licensed continuation schema in a redescription-neutral fixed-domain evaluative continuity problem must carry a redescription action on transition labels and satisfy*

$$R(T(X)) \sim (R_{\#}T)(R(X)).$$

Proof. A continuation schema is evaluatively licensed only if the result does not depend on an arbitrary choice of presentation. Applying a redescription to the input may also redescribe the continuation label or schema; otherwise the label would function as a hidden parameter. Therefore there must be an induced action $R_{\#}$ on continuation schemas. Equality of the two orders—continue then redescribe, or redescribe then continue with the transported schema—is precisely the covariance equation. Failure of this equation means the continuation is presentation-dependent and is not licensed in a fixed-domain neutral regime. $\square$

Lemma 3.8 (Same-regime continuity forces binary conservation). *In a non-degenerate fixed-domain continuity problem, licensed same-regime continuation must preserve the binary participation status. A map that promotes, repairs, or destroys participation is a boundary-crossing, termination, reset, or violation rather than a licensed same-regime continuation.*

Proof. A same-regime continuation is supposed to track continuity inside the same evaluative domain. If it maps a positive participant to a negative one, the asserted continuity has terminated or failed. If it maps a negative participant to a positive one, it has introduced a repair, reset, or new source of participation not licensed by the prior same-domain status. Because the status is binary and non-degenerate, either direction changes the regime-level participation verdict. Hence licensed same-regime continuation preserves the predicate; non-preserving maps are modeled outside the licensed transition class. $\square$

Lemma 3.9 (Internality forces parameter exclusion). *If a mechanism consults data not represented in the state, witness basis, or transported transition-label structure, then the mechanism is not fixed-domain redescription-neutral. It has introduced an external parameter.*

Proof. A datum invisible to redescription transformations cannot be transported with the represented state or with the transition schema. A choice depending on that datum therefore may distinguish presentations that the fixed-domain witness structure cannot distinguish, or may select between symmetric alternatives without an invariant basis. In either case the datum is external to the fixed domain. Redescription-neutral evaluative continuity therefore forces internal data closure. $\square$

Lemma 3.10 (Finite continuity witnesses force grounded transport). *A positive continuity claim that has neither a non-derivative source nor a well-founded predecessor profile lacks a finite invariant continuity witness. It is therefore either an infinite regress, a stipulative label, or a different regime with an explicit reset/source operation.*

Proof. A finite continuity witness must exhibit either a finite predecessor profile ending in a source or a certified same-domain continuation from such a source. If every positive state depends on an earlier positive state with no well-founded base, then no finite extraction terminates in a source. The remaining options are to accept an infinite regress without finite witness extraction, to assign the positive label independently of transport, or to introduce a reset/source operation. The first violates finite witness adequacy; the second is stipulative; the third changes the regime description. Thus grounded transport is forced for determinate same-domain continuity. $\square$

Theorem 3.11 (Evaluative inevitability theorem). *Every non-degenerate fixed-domain evaluative continuity problem satisfying continuity adequacy canonically restricts to a grounded fixed-domain admissible evaluation system*

$$\mathcal{M} = (S, \sim, \mathcal{E}, \mathcal{R}, \mathcal{T}, \text{Elig})$$

up to consequence-equivalent witness choice and admissible conjugacy. Consequently, the quotient interface, descent structure, symmetry obstruction, normal-form exhaustion, non-duplication result, role decomposition, and local rigidity theorem are not optional additions; they are the forced reduced-interface consequences of determinate fixed-domain evaluative continuity.

Proof. By Lemma 3.5, witness determinacy forces a coherent evaluation functional $\mathcal{E} : S \rightarrow \Sigma$. By Lemma 3.6, presentation neutrality forces the admissible presentation transformations to form the redescription groupoid used above. By Lemma 3.7, stable continuation forces the $R_{\#}$ covariance structure on licensed transitions. By Lemma 3.8, same-regime continuity forces binary conservation of the participation predicate, which becomes Elig. By Lemma 3.9, fixed-domain neutrality forces internal data closure and excludes hidden parameters. By Lemma 3.10, finite continuity witnessing forces grounded transport. These are exactly the structural clauses of a grounded fixed-domain admissible evaluation system, modulo deletion of redundant witnesses and harmless recoding covered by Lemma 2.4 and Theorem 2.6. Applying the later quotient, persistence, symmetry, exhaustion, non-duplication, decomposition, and rigidity theorems gives the stated forced consequences. $\square$

Corollary 3.12 (Constraint architecture forces kernel instantiation). *A non-degenerate fixed-domain evaluative continuity problem satisfying continuity adequacy cannot instantiate coherent evaluability without instantiating the kernel, up to consequence-equivalent witness choice, harmless signature recoding, and admissible conjugacy. A purported same-domain alternative that avoids the kernel must fail one of the preservation demands or leave the regime.*

Proof. Theorem 3.11 forces the witness-coherent evaluation, redescription, covariance, eligibility, internality, and grounding clauses. The remaining kernel clauses are supplied by the AASC closure imports in Section 4: boundary non-selection, standing/admissibility identity, operator-envelope exhaustion, no-hidden-remainder, role exhaustion, and retained-trajectory closure. If any such clause is absent, the corresponding proof identifies the failure as a hidden parameter, unsupported selector, competing gate, inadmissible operator, bookkeeping-only surplus, failed standing, or explicit regime change. Hence kernel avoidance is not a distinct same-domain coherent alternative. $\square$

Remark 3.13 (Closure is emergent, not imposed). *The closure structure is not imposed axiomatically as a desired conclusion. It is the common residue of five preservation demands: determinate witness extraction, redescription neutrality, schema-stable continuation, internal data dependence, and finite continuity grounding. Removing any one demand permits a familiar failure mode: arbitrary signatures, presentation-dependent verdicts, hidden parameters, ungrounded continuity, or regime-changing repair.*

4 AASC Full-Import Closure Layer

The preceding inevitability theorem forces the core admissible evaluation structure from determinate fixed-domain continuity demands. This section imports the remaining AASC closure tools needed to make the comparison class fully closed: boundary fixation, AMetric non-selection, standing/admissibility gate identity, same-scope operator exhaustion, no-hidden-remainder, and coherent retained-trajectory closure. These results are stated in the same notation as the present paper so that the later quotient, branch, persistence, and rigidity theorems do not depend on untranslated framework vocabulary.

Definition 4.1 (Evaluated fragment and boundary gate). *An evaluated fragment is a subset $E \subseteq S$ on which fixed-domain eligibility is actually assessed. A boundary gate is a partial map*

$$\text{Gate} : E \rightarrow \{0, 1\}$$

such that $\text{Gate}(X) = \text{Elig}(X)$ for every $X \in E$. Boundary-gate work is classificatory: it settles whether a represented state belongs to the standing-positive side of the fixed domain. It is not an interior operator, trajectory step, optimization routine, or selector.

Definition 4.2 (Boundary-authoritative datum). *A datum d is boundary-authoritative when changing d can change boundary typing, evaluated-fragment membership, Gate, Elig, $\mathcal{E}$, the admissible quotient, the quotient transition relation, the predecessor relation, or the branch-resolution status of a same-domain problem. Boundary-authoritative data must be represented in $\mathbf{W}$ or in the fixed transition-label structure and must be acted on by $\mathcal{R}$ and $R_\#$.*

Definition 4.3 (AMetric boundary discipline). *The AMetric boundary is the interface at which admissibility-bearing participation is classified before any interior operator is available. At that boundary, no selector, grading, order, metric, probability, typicality rule, canonical representative, discrete index, or optimization parameter has admissibility-bearing authority unless it is already fixed as a witness, anchor, declared boundary trace, or transition-label invariant of the evaluated domain. Post-fixation structure may be used as interior bookkeeping after eligibility has been fixed, but it may not be transported backward to decide the boundary classification it presupposes.*

Lemma 4.4 (AMetric non-selection). *At the AMetric boundary, a same-domain rule cannot select one representative from a nontrivial redescription orbit by appeal to a metric, score, ranking, probability, typicality, order, canonical representative, or discrete index unless the selecting datum is already represented as an admissible invariant. If it is not represented, the rule is an external parameter or a scope change; if it is represented, the case is ordinary invariant asymmetry rather than boundary selection.*

Proof. Before boundary fixation, the only authoritative information is the represented witness and envelope structure preserved by admissible redescription. A metric, rank, score, index, or representative convention that is not represented in this structure is not acted on by $\mathcal{R}$, so using it distinguishes states by a datum outside the fixed domain. That is parameter injection or scope change. If the datum is represented and transported, it is no longer a boundary-side selector; it is a same-domain invariant and falls under the invariant-asymmetry class already tracked by the theory. $\square$

Definition 4.5 (Declared boundary trace). *For a role-assignment or branch-fixation problem with candidate set $C \subseteq S$, a declared boundary trace is a witness-extractable partial predicate*

$$\text{Tr} : C \rightarrow \{0, 1\}$$

that is included in the fixed domain before the role or branch is used, is invariant under channel-preserving admissible redescriptions, and is fail-closed when undefined, non-singleton, or unstable. A boundary trace is not an environmental oracle: it is a bivalent status trace represented inside the target evaluation regime.

Theorem 4.6 (Boundary-trace fixation). *Consider a same-domain role or branch-fixation problem whose internally definable invariants are neutral on a candidate set C , and whose declared indexical, provenance, or origin markers are absent or non-singleton on C . If a singleton stable declared boundary trace exists, then it supplies the unique admissible asymmetry fixing the role or branch occupant. If no such trace exists, then no same-domain admissible selector fixes a unique occupant.*

Proof. Any same-domain selector must draw authority from one of the locations available inside the fixed domain. If the decisive feature is internally definable, it is an internal invariant. If it

is an origin, position, or production-history marker, it is an indexical/provenance invariant. If it is neither internal nor provenance-based but remains declared in the target domain, the only remaining admissible location is a boundary trace. All other deciding features are preferences, scores, salience facts, post-selection outcomes, or external metadata and are ruled out by Lemma 4.4 and internal data closure. A singleton stable trace is preserved by the admissible transformations that preserve the channel package, so it fixes the unique occupant. Absence, multiplicity, instability, or undefinedness leaves no admissible selector. $\square$

Corollary 4.7 (No post-selection boundary repair). *A boundary trace may fix a role only if it is declared before role use. A trace introduced after a failed or ambiguous selection is either an explicit scope reset or an inadmissible repair.*

Proof. If the trace is added after selection failure, the original fixed domain did not contain the selecting datum. Treating the later trace as same-domain authority imports a parameter invisible to the original $\mathcal{R}$ -action. Declaring it openly changes the domain; leaving it undeclared is repair or selector smuggling. $\square$

Definition 4.8 (Same-domain admissibility classifier). *A map $A : S \rightarrow \{0, 1\}$ is a same-domain admissibility classifier when it purports to govern the same evaluated fragment, quotient identifications, licensed continuations, and retained-history structure governed by Elig . It is genuinely distinct from eligibility when there exists $X \in S$ with $A(X) \neq \text{Elig}(X)$ or when it agrees extensionally but claims different same-domain governance authority.*

Theorem 4.9 (Standing-admissibility identity closure). *No same-domain admissibility classifier genuinely distinct from Elig survives in a closed fixed-domain admissible evaluation system. Any attempted separation between admissibility and standing must instantiate at least one of the following routes:*

- (G1) *standing-neutral admissibility;*
- (G2) *deferred admissibility or deferred standing on a retained lineage;*
- (G3) *admissibility reset after failure;*
- (G4) *factorized same-domain governance by two independent gates;*
- (G5) *illicit modification of standing by repair, parameter injection, or scope transport.*

Each route collapses to bookkeeping, explicit regime change, or violation of fail-closed admissibility. Hence $A(X) = \text{Elig}(X)$ for every evaluated state and the two gates have the same governance role on the fixed domain.

Proof. If A and Elig differ on an evaluated state, the mismatch is either $(A, \text{Elig}) = (1, 0)$ or $(0, 1)$. In the first case admissibility is asserted without standing, so either it has no governance effect, which is bookkeeping, or it is deferred to later standing, which violates retained-lineage fail-closedness. In the second case standing is asserted without admissibility, so the regime either resets a negative classification, splits governance into a second gate, or modifies standing outside the certified boundary. These are routes (G3)–(G5). If the predicates agree extensionally but are said to govern differently, the difference must alter evaluated-state classification or retained trajectory behavior; otherwise it is notation. Altering either creates a second same-domain gate or hidden reset. Since each route is ruled out by boundary non-selection, fail-closedness, redescription neutrality, and internal data closure, no genuine separation remains. $\square$

Corollary 4.10 (No competing gates). *Within the fixed domain, there is no second admissibility-bearing gate below, beside, or above Elig. A purported second gate either factors through Elig, changes the regime, or violates fixed-domain closure.*

Definition 4.11 (Certified operator envelope). *A certified same-scope operator envelope is a tuple*

$$\text{Op} = (D_{\text{Op}}, F_{\text{Op}}, \text{Inv}_{\text{Op}}, \text{Fail}_{\text{Op}}, \text{Rev}_{\text{Op}}, \text{cert}_{\text{Op}})$$

where $D_{\text{Op}} \subseteq S$ is the domain of definition, $F_{\text{Op}} : D_{\text{Op}} \rightarrow S$ is a partial operator, Inv_{Op} records the witnesses and invariants preserved by the operator, Fail_{Op} records the failure and undefinedness conditions, Rev_{Op} records the revocation conditions under which the operator is no longer same-scope, and cert_{Op} is an independent admissibility certificate. Operators are interior envelopes: they may act only after the relevant boundary classification and domain conditions have already been fixed.

Definition 4.12 (Bookkeeping-equivalent operator). *A certified operator is bookkeeping-equivalent when, on every eligible input in its envelope, it preserves eligibility, witness consequences, the admissible quotient class, quotient transition consequences, and predecessor profile. It may change notation, presentation, or internal accounting, but it adds no new admissibility-bearing power on the same fixed domain.*

Theorem 4.13 (Same-scope operator-envelope exhaustion). *Every same-scope operator used to select, complete, totalize, force, saturate, repair, canonicalize, diagonalize, self-apply, invoke a fixed point, appeal to a universal property, change foundations, or transfer structure is either bookkeeping-equivalent on the fixed domain or inadmissible as a same-domain operator. In particular, no same-scope operator creates a new admissibility gate, total evaluator, hidden kernel, or standing-repair mechanism.*

Proof. Let F be a purported same-scope operator. If all data on which F depends are already witness-fixed and F preserves the eligibility, quotient, transition, and predecessor profile, then F adds no new admissibility-bearing distinction and is bookkeeping-equivalent. Otherwise F must do at least one of the following: enlarge the domain, collapse distinct carriers, complete a partial evaluator, import a family-specific datum, repair an ineligible or undefined input, or change the operator family or foundation. Domain enlargement and foundation change are scope changes; carrier collapse and canonicalization alter the quotient or reduce to a declared section; imported data are external parameters; repair and totalization violate fail-closedness; diagonal, self-referential, fixed-point, forcing, saturation, compactness, and universal-property detours require either an already fixed witness profile or a completed comparison totality. In the first case they are bookkeeping; in the second they import structure not present in the fixed domain. The listed cases exhaust the ways an operator can have same-domain force, so no hidden same-scope operator remains. $\square$

Corollary 4.14 (No same-domain diagonal escape). *A same-domain diagonal, self-reference, fixed-point, or universality maneuver cannot generate a new admissible interface layer. It either reports a structure already fixed by the quotient interface or leaves the fixed domain by importing an evaluator, completion, or meta-level authority.*

Definition 4.15 (Governance-bearing locus). *A same-domain variation is governance-bearing when it can change at least one of the following:*

- (L1) *boundary typing or evaluated-fragment membership;*
- (L2) *the standing/admissibility partition, equivalently Elig by Theorem 4.9;*

- (L3) the witness-coherent evaluation signature or boundary trace package;
- (L4) the lawful quotient induced by admissible redescription;
- (L5) admissible continuation, predecessor, or coherent retained-trajectory structure;
- (L6) anchor–tensor–skin role assignment on the fixed carrier.

Theorem 4.16 (Fixed-domain no-hidden-remainder). *Within the fixed comparison class, every same-domain admissibility-bearing variation changes at least one governance-bearing locus. If it changes none, it is bookkeeping or skin. If it changes something outside the list, then it changes the domain, witness basis, boundary package, operator family, or admissibility regime.*

Proof. A variation that alters evaluated membership or boundary typing falls under (L1). If it changes which states are standing-positive, it falls under (L2). If it changes the represented consequences or a boundary trace, it falls under (L3). If it preserves these but separates or merges states differently under lawful redescription, it changes the quotient and falls under (L4). If it changes possible continuation, predecessor, or retained history behavior, it falls under (L5). If it changes the admissibility-bearing work done by a descriptive element, it changes the role assignment and falls under (L6). A variation that changes none of these has no admissibility-bearing effect in the fixed comparison class and is bookkeeping. Any alleged further effect requires an unlisted datum or operation; by internal data closure and operator-envelope exhaustion, that is a scope change or inadmissible import. $\square$

Corollary 4.17 (No fourth same-domain role). *There is no admissibility-preserving fourth role beyond anchor, tensor, and skin on the same fixed carrier. A proposed fourth role changes eligibility, changes quotient/transport/trajectory structure without changing eligibility, changes neither, or changes scope; these are respectively anchor, tensor, skin, or regime change.*

Definition 4.18 (Audit trajectory and coherence). *An audit trajectory is a finite nonempty sequence $\tau = (X_0, \dots, X_n)$ whose adjacent steps are candidate same-domain continuation attempts. It is coherent, written $\text{Coh}(\tau)$, when every retained state on the sequence is eligible and every adjacent step is certified by a same-domain envelope. A failure witness is an index $i \leq n$ with $\text{Elig}(X_i) = 0$ or with an uncertified adjacent step entering or leaving X_i .*

Theorem 4.19 (No deferred coherence). *If an audit trajectory contains an incoherent prefix or a failure witness, no admissible same-domain continuation restores coherence while retaining that prefix. Any later positive-looking continuation is either a new trajectory after reset, a scope change, or post hoc repair.*

Proof. A retained prefix is not deleted by later continuation. If it contains an ineligible state or uncertified step, the same witness remains present in every extension that retains the prefix. Treating the later suffix as restoring the whole would require a repair of an already failed retained lineage or a reset of the identity of the evaluated trajectory. Fail-closedness and Corollary 4.7 exclude the first; explicit reset is a different regime. $\square$

Theorem 4.20 (Constructional behavior exhaustion). *In an AASC-closed fixed-domain evaluation system, governance-relevant constructional behavior is exhausted by two loci: classification of evaluated states by the boundary gate and coherent retained-trajectory structure. Consequently, evaluative continuity is not an extra metaphysical premise; it is the retained-history locus forced once the fixed-domain construction problem includes same-domain continuation.*

Proof. A same-domain constructional behavior either concerns whether an evaluated state is admitted into the standing-positive side, or concerns how an already retained history continues. The first is boundary-gate classification. The second is retained-trajectory structure. A proposed third behavior must leave both the state classification and retained continuation structure unchanged while still doing governance-bearing work; this contradicts Theorem 4.16. Thus continuity appears because same-domain construction has retained-history structure, not because a metaphysical continuity primitive was added. $\square$

Theorem 4.21 (AASC full-import closure theorem). *A non-degenerate fixed-domain evaluative continuity problem satisfying continuity adequacy and the AASC closure imports above has no surviving same-domain escape by boundary selector, metric parameter, standing/admissibility separation, second gate, hidden operator, same-domain diagonalization, role refinement, post-selection repair, or hidden governance-bearing remainder. Every purported escape reduces to invariant asymmetry, declared boundary trace, canonical section, bookkeeping, explicit regime change, parameter injection, or fail-closed violation.*

Proof. Boundary selectors, metrics, scores, indices, and rankings are closed by Lemma 4.4. Role or branch fixation by the last admissible asymmetry source is closed by Theorem 4.6. Separation of standing and admissibility, including competing gates, is closed by Theorem 4.9 and Corollary 4.10. Operator totalization, repair, completion, diagonalization, fixed-point, forcing, and universality detours are closed by Theorem 4.13 and Corollary 4.14. Role refinements are closed by Corollary 4.17. Post-selection and trajectory repair are closed by Corollary 4.7 and Theorem 4.19. Any remaining alleged effect must be a governance-bearing variation, but Theorem 4.16 forces it into one of the listed loci. Therefore no further same-domain escape remains. $\square$

Corollary 4.22 (No permissive-alternative reading). *The AASC-closed comparison class is not a permissive taxonomy of rival same-domain evaluation regimes. Within the fixed-domain non-degenerate coherence problem, a structure that lacks the kernel is not an alternative admissible architecture; it is an under-specified boundary, a hidden parameterization, a competing gate, an illicit operator envelope, a repair attempt, a role conflation, bookkeeping, or a regime change.*

Proof. This is the contrapositive reading of Theorem 4.21 together with Corollary 3.12. The full-import closure theorem exhausts same-domain escape loci, and the kernel-instantiation corollary identifies the minimal surviving constraint package. Therefore anything outside the package is classified by its failure mode rather than by an additional admissible same-domain architecture. $\square$

5 Projection as Quotient Descent

5.1 Admissible orbit equivalence

Definition 5.1 (Admissible orbit equivalence). *For $X, Y \in S$, define*

$$X \approx Y \iff X \equiv_{\text{orb}} Y \iff \exists R \in \mathcal{R} \text{ such that } R(X) \sim Y.$$

Let $[X]$ denote the $\approx$ -equivalence class of X , and let

$$\pi : S \rightarrow S/\approx, \quad \pi(X) = [X]$$

be the quotient map. Thus π removes admissible presentation-orbit variation, not all possible semantic or observational distinctions.

Lemma 5.2 (Orbit equivalence is an equivalence relation). *The relation $\approx$ is reflexive, symmetric, and transitive.*

Proof. Reflexivity follows from $\text{id}_S \in \mathcal{R}$ and $X \sim X$. If $X \approx Y$, choose $R \in \mathcal{R}$ with $R(X) \sim Y$. By the inverse clause for $\mathcal{R}$, there is $R^{-1} \in \mathcal{R}$ with $R^{-1}(Y) \sim R^{-1}R(X) \sim X$, so $Y \approx X$. If $X \approx Y$ and $Y \approx Z$, choose $R_1, R_2 \in \mathcal{R}$ with $R_1(X) \sim Y$ and $R_2(Y) \sim Z$. Since R_2 preserves $\sim$ and $R_2 \circ R_1 \in \mathcal{R}$,

$$(R_2 \circ R_1)(X) \sim R_2(Y) \sim Z,$$

so $X \approx Z$. □

Lemma 5.3 (Eligibility descends to the quotient). *There is a unique map*

$$\overline{\text{Elig}} : S/\approx \rightarrow \{0, 1\}$$

such that $\overline{\text{Elig}}([X]) = \text{Elig}(X)$.

Proof. If $[X] = [Y]$, then $X \approx Y$. Choose $R \in \mathcal{R}$ with $R(X) \sim Y$. Redescription neutrality gives $\text{Elig}(R(X)) = \text{Elig}(X)$, and $\sim$ -invariance gives $\text{Elig}(R(X)) = \text{Elig}(Y)$. Hence $\text{Elig}(X) = \text{Elig}(Y)$, so the definition is independent of representative. Uniqueness follows from surjectivity of π . □

Lemma 5.4 (Evaluation descends to the quotient). *There is a unique map*

$$\overline{\mathcal{E}} : S/\approx \rightarrow \Sigma$$

such that $\overline{\mathcal{E}}([X]) = \mathcal{E}(X)$.

Proof. If $[X] = [Y]$, choose $R \in \mathcal{R}$ with $R(X) \sim Y$. Redescription admissibility gives $\mathcal{E}(R(X)) = \mathcal{E}(X)$. Since $\sim$ is identity-preserving, $\mathcal{E}(R(X)) = \mathcal{E}(Y)$. Hence $\mathcal{E}(X) = \mathcal{E}(Y)$. Surjectivity of π gives uniqueness. □

5.2 Transition descent

Definition 5.5 (Quotient transition relation). *The quotient transition relation of $\mathcal{M}$ is the relation*

$$\widehat{\mathcal{T}}_Q \subseteq (S/\approx) \times (S/\approx)$$

defined by

$$[X] \widehat{\mathcal{T}}_Q [Y] \iff \exists T \in \mathcal{T} \text{ such that } Y \approx T(X).$$

If a transition schema T is redescription-stable, one may also consider the individual quotient map $\widehat{T}$ defined below.

Lemma 5.6 (Transition covariance preserves quotient transition structure). *The relation $\widehat{\mathcal{T}}_Q$ is independent of representatives. More precisely, if $X' \approx X$ and $[X] \widehat{\mathcal{T}}_Q [Y]$, then there exists Y' with $Y' \approx Y$ and $[X'] \widehat{\mathcal{T}}_Q [Y']$.*

Proof. Choose $T \in \mathcal{T}$ with $Y \approx T(X)$. Since $X' \approx X$, choose $R \in \mathcal{R}$ with $R(X) \sim X'$. Transition covariance gives

$$R(T(X)) \sim (R_{\#}T)(R(X)) \sim (R_{\#}T)(X').$$

Because $R(T(X)) \approx T(X) \approx Y$, the state $Y' = (R_{\#}T)(X')$ is quotient-equivalent to Y . Since $R_{\#}T \in \mathcal{T}$, we have $[X'] \widehat{\mathcal{T}}_Q [Y']$. Thus the quotient transition relation is well-defined on classes. □

Corollary 5.7 (Stable individual transitions descend). *If $T \in \mathcal{T}$ is redescription-stable, then there is a unique induced map*

$$\widehat{T} : S/\approx \rightarrow S/\approx$$

such that $\widehat{T}([X]) = [T(X)]$ for every $X \in S$.

Proof. If $X' \approx X$, the proof of Lemma 5.6 gives

$$R(T(X)) \sim (R_{\#}T)(R(X)) = T(R(X)) \sim T(X'),$$

so $T(X') \approx T(X)$. Hence $[T(X)]$ is independent of representative. Uniqueness follows from surjectivity of the quotient map. $\square$

Theorem 5.8 (Quotient descent). *Every licensed transition schema descends to the quotient transition relation $\widehat{T}_Q$. Every redescription-stable licensed transition schema descends further to a unique quotient map $\widehat{T}$ satisfying*

$$\pi \circ T = \widehat{T} \circ \pi.$$

Proof. The first claim is Lemma 5.6. The second claim is Corollary 5.7; the displayed factorization is the definition of $\widehat{T}$. $\square$

Corollary 5.9 (Projection necessity). *The minimal presentation-independent interface through which licensed same-domain evaluation passes is the quotient projection $\pi : S \rightarrow S/\approx$ together with the descended evaluation functional $\overline{\mathcal{E}}$, descended eligibility $\overline{\text{Elig}}$, the quotient transition relation $\widehat{T}_Q$, and the induced maps $\widehat{T}$ for redescription-stable transition schemas.*

Proof. By Theorem 5.8, all licensed transition schemas induce well-defined quotient-level transition structure, and stable schemas induce quotient maps. Any interface that identifies fewer states than $\approx$ retains presentation degrees of freedom that do no admissibility-relevant work; any interface that identifies more states can merge eligibility-, witness-, or transition-relevant distinctions. Thus the quotient is the canonical minimal projection layer for the fixed system. $\square$

Proposition 5.10 (Minimal quotient interface universal property). *Let J be any fixed-domain representation with a map $\eta : S \rightarrow J$ such that:*

- (U1) $\eta(RX) = \eta(X)$ for every admissible redescription $R \in \mathcal{R}$;
- (U2) η preserves descended eligibility and witness-derived evaluation signatures; and
- (U3) whenever $[X]\widehat{T}_Q[Y]$, the corresponding η -images carry the represented transition consequence in J .

Then there exists a unique map $\bar{\eta} : S/\approx \rightarrow J$ with

$$\eta = \bar{\eta} \circ \pi.$$

If J also separates all admissibility-relevant invariant tests, then $\bar{\eta}$ is faithful on the reduced interface. Thus any admissibility-preserving presentation-independent representation factors through the quotient interface.

Proof. Define $\bar{\eta}([X]) = \eta(X)$. If $[X] = [Y]$, then $Y \sim R(X)$ for some $R \in \mathcal{R}$. Since η is redescription-invariant and respects immediate presentation identity, $\eta(Y) = \eta(RX) = \eta(X)$; hence $\bar{\eta}$ is well-defined. The equation $\eta = \bar{\eta} \circ \pi$ follows directly, and uniqueness follows because π is surjective. If J separates all admissibility-relevant invariant tests, two quotient classes with distinct reduced-interface profiles cannot have the same J -image, so $\bar{\eta}$ is faithful on the reduced interface. $\square$

6 Persistence from Well-Founded Eligible Transport

6.1 Origins and derivative bearers

Definition 6.1 (Derivative and non-derivative eligible state). *Let $\mathcal{M}$ be grounded with strict eligible predecessor relation $\prec$. A state $Y \in S^+$ is derivative if there exists $X \in S^+$ with $X \prec Y$. It is non-derivative if no such X exists. A non-derivative eligible state is also called an origin.*

Definition 6.2 (Transport witness). *A transport witness for an eligible state Y is a terminating backward certificate showing how Y is grounded by strict eligible predecessor descent. Concretely, it is either a finite predecessor chain ending at an origin or, in the transfinite case, a well-founded rank assignment that strictly decreases to every predecessor.*

Theorem 6.3 (Transport grounding trichotomy). *Let $S^+ \neq \emptyset$ and let $\prec$ be a redescription-invariant strict eligible predecessor relation. If $\prec$ is not well-founded, then any claimed persistence or singular-continuity assignment determined by $\prec$ faces at least one of the following alternatives:*

- (i) *finite or well-founded transport auditing fails for some nonempty positive region;*
- (ii) *the continuity assignment is made without a terminating transport witness and is therefore stipulative relative to the fixed-domain transport structure;*
- (iii) *no invariant continuity witness exists for that region.*

Consequently, any non-stipulative fixed-domain persistence regime with auditable continuity witnesses must use well-founded eligible transport.

Proof. If $\prec$ is not well-founded, there exists a nonempty subset $A \subseteq S^+$ with no $\prec$ -minimal element. Thus every $a \in A$ has a predecessor $b \in A$ with $b \prec a$. Backward predecessor descent inside A therefore never reaches an origin. Hence a terminating finite certificate does not exist for A , and no well-founded rank can be assigned on A with strict decrease to predecessors, since such a rank would force a minimal-rank element of A . This proves (i).

Now suppose a persistence or singular-continuity predicate is nevertheless assigned on A . If the assignment is accepted despite the absence of any terminating transport witness, then its value is not determined by the fixed-domain predecessor structure; it is a stipulative label relative to that structure, giving (ii). If the assignment is not so accepted, then the region has no invariant continuity witness, giving (iii). The final sentence follows by contraposition: a non-stipulative auditable persistence regime must exclude the non-well-founded case. $\square$

Lemma 6.4 (Infinite descent collapse). *Let $A \subseteq S^+$ be a nonempty positive region with no $\prec$ -minimal element. Then no finite invariant continuity-witness extraction procedure can certify every element of A from within A .*

Proof. Take any $a_0 \in A$. Because A has no minimal element, there is $a_1 \in A$ with $a_1 \prec a_0$; again there is $a_2 \in A$ with $a_2 \prec a_1$, and so on. Any finite backward extraction beginning at a_0 either stops at some $a_n \in A$, in which case a_n still has a predecessor in A and is not an origin, or it leaves A , in which case the witness is not extracted from the region's own transport structure. Therefore a finite same-region certificate cannot be supplied. Since $\prec$ is redescription-invariant, admissible redescription transports the same failure across the orbit of the region; the obstruction is invariant rather than a presentation artifact. $\square$

Lemma 6.5 (Minimal elements in well-founded transport). *If $(S^+, \prec)$ is well-founded and $A \subseteq S^+$ is nonempty, then A contains an element a such that no $b \in A$ satisfies $b \prec a$.*

Proof. This is exactly the defining property of well-foundedness. $\square$

Theorem 6.6 (Origin necessity). *Every grounded fixed-domain admissible evaluation system has at least one non-derivative eligible state.*

Proof. By non-degeneracy, $S^+ \neq \emptyset$. Apply Lemma 6.5 to $A = S^+$. The resulting minimal $O \in S^+$ has no eligible predecessor, hence is non-derivative. $\square$

Corollary 6.7 (No eligible infinite regress). *There is no infinite sequence*

$$Y_0 \succ Y_1 \succ Y_2 \succ \dots$$

of eligible states, where $Y_{n+1} \prec Y_n$ for every n .

Proof. An infinite descending sequence contradicts well-foundedness of $\prec$. $\square$

6.2 Rank without the shortest-chain error

The earlier finite-rank argument is replaced by the standard well-founded rank construction. This avoids the invalid assumption that an arbitrary predecessor of Y must lie on a shortest origin-chain to Y .

Definition 6.8 (Well-founded transport rank). *For $Y \in S^+$, define the ordinal rank*

$$\rho(Y) = \sup\{\rho(X) + 1 : X \prec Y\}.$$

If Y has no predecessor, the supremum of the empty set is 0.

Lemma 6.9 (Rank strictly decreases to predecessors). *If $X \prec Y$, then $\rho(X) < \rho(Y)$.*

Proof. By definition,

$$\rho(Y) = \sup\{\rho(Z) + 1 : Z \prec Y\}.$$

Since $X \prec Y$, the term $\rho(X) + 1$ occurs in the displayed set. Hence $\rho(X) + 1 \leq \rho(Y)$, so $\rho(X) < \rho(Y)$. $\square$

Corollary 6.10 (Finite backward walks terminate). *If one iteratively chooses a strict eligible predecessor whenever one exists, the process cannot continue through an infinite sequence. Every such backward walk terminates at an origin.*

Proof. An infinite backward walk would contradict Corollary 6.7. A terminal point of a backward walk is precisely an eligible state with no strict eligible predecessor, i.e. an origin. $\square$

Remark 6.11 (What persistence means here). *Persistence is not introduced as a primitive mental fact. It is the well-founded continuation structure of eligible states under licensed same-domain transport. The non-derivative origin is forced by groundedness and non-vacuity, not by stipulating a special bearer.*

7 Symmetric Branching and Equivariant Choice Failure

7.1 Branch points and local symmetry

Definition 7.1 (Branch point). *A state $X \in S$ is a two-branch point if there exist $T_1, T_2 \in \mathcal{T}$ such that*

$$Y_1 = T_1(X), \quad Y_2 = T_2(X), \quad Y_1 \not\sim Y_2.$$

The outputs are presentation-distinct. They may nevertheless lie in the same admissible presentation orbit when the branch is symmetric; in that case the quotient records the orbit but does not select a representative. The branch is irreversible when no licensed same-domain transition returns either branch to X up to the relevant branch-history structure.

Definition 7.2 (Local branch symmetry). *A two-branch point X with branch outputs $Y_1 = T_1(X)$ and $Y_2 = T_2(X)$ is symmetric when there exists $R_{\leftrightarrow} \in \mathcal{R}$ such that*

$$R_{\leftrightarrow}(X) \sim X, \quad R_{\leftrightarrow}(Y_1) \sim Y_2, \quad R_{\leftrightarrow}(Y_2) \sim Y_1,$$

and, at the transition-schema level, $R_{\leftrightarrow\#}$ exchanges the two branch labels up to admissible label equivalence. Equivalently, the stabilizer of X under admissible redescription acts transitively on the two branch alternatives. This is compatible with transition covariance: a branch label may be carried to its symmetric counterpart rather than fixed pointwise.

Definition 7.3 (Finite-support renaming richness). *A branch point satisfies finite-support renaming richness when:*

- (N1) *all relevant states have finite typed name support;*
- (N2) *each type has an infinite fresh-name supply;*
- (N3) *admissible redescrptions include capture-avoiding type-preserving renamings of finite supports; and*
- (N4) *the local envelope contains enough unused presentation names to carry a nontrivial swap without changing anchor data.*

Lemma 7.4 (Sufficient condition for branch symmetry). *If a two-branch point has finite-support renaming richness and no admissible-invariant predicate distinguishes the two branch outputs, then it is symmetric.*

Proof. Let $A = \text{supp}(Y_1)$ and $B = \text{supp}(Y_2)$ be the finite typed supports of the branch outputs. Fresh-name supply permits alpha-renaming, if necessary, so that the relevant supports can be made disjoint without changing any anchor or eligibility datum. Choose a type-preserving bijection between the finite supports and extend it to a total capture-avoiding bijection of the infinite typed name supply. This total renaming is admissible by richness. Because no admissible-invariant predicate distinguishes the branches, the resulting redescription is not blocked by fixed-domain structure. It fixes X up to $\sim$ and swaps Y_1, Y_2 up to $\sim$. $\square$

7.2 Equivariant selectors

Definition 7.5 (Branch selector). *At a two-branch point X with outputs Y_1, Y_2 , a singular branch selector is a partial map $C : S \rightarrow S$ satisfying*

$$C(X) \in \{Y_1, Y_2\}.$$

It is required to respect base presentation identity: if $Z \sim Z'$ and both sides are defined, then $C(Z) \sim C(Z')$. It is equivariant when, for every admissible redescription R for which both sides are defined,

$$C(R(Z)) \sim R(C(Z)).$$

Lemma 7.6 (Internal stable mechanisms are equivariant). *Any selector computed only from internal fixed-domain data in a redescription-neutral covariant system is equivariant.*

Proof. Internal data closure says that every datum consulted by the selector is part of the state or fixed transition signature and is acted on by admissible redescription. Transition covariance and redescription neutrality say that applying R before or after the computation preserves the identity-relevant result up to $\sim$, with transition labels carried by $R_{\#}$ when labels are involved. Therefore the selector commutes with admissible redescription up to $\sim$, which is equivariance. $\square$

Theorem 7.7 (No equivariant selector at a symmetric branch). *At a symmetric two-branch point X with presentation-distinct outputs Y_1, Y_2 , no equivariant singular branch selector exists.*

Proof. Assume C is an equivariant singular selector. Since $C(X)$ is one of Y_1, Y_2 , suppose without loss of generality that $C(X) = Y_1$ up to $\sim$. Let $R_{\leftrightarrow}$ be the branch-swapping redescription. Because $R_{\leftrightarrow}(X) \sim X$, equivariance gives

$$C(X) \sim C(R_{\leftrightarrow}(X)) \sim R_{\leftrightarrow}(C(X)) \sim R_{\leftrightarrow}(Y_1) \sim Y_2.$$

Thus $Y_1 \sim Y_2$, contradicting the presentation-distinct branch-output assumption. The case $C(X) = Y_2$ is identical. $\square$

Corollary 7.8 (Symmetric self-location impossibility). *At a symmetric irreversible branch point, a singular representative-selection verdict cannot be produced by an internal admissible mechanism. A successful verdict must either use invariant asymmetry, add a canonical section, inject an external parameter, or leave the fixed-domain regime.*

Proof. By Lemma 7.6, an internal admissible mechanism would be equivariant. Theorem 7.7 excludes such selectors at symmetric presentation-distinct branches. The listed alternatives are classified in Section 8. $\square$

7.3 Worked toy model: symmetric representative choice

This example is not a model of psychology or consciousness. It is a minimal finite model showing why the selector theorem is not merely verbal. The toy model is illustrative rather than exhaustive; none of the core results depends on finite state cardinality, exact discrete symmetry, or explicit naming representations. Infinite state spaces, continuous state spaces, or richer automorphism groups only replace the two-point swap below by a larger admissible orbit action. The toy construction is illustrative only and carries no privileged ontological or representational status.

Let

$$S = \{x, a, b, f\}, \quad \sim = \text{equality}.$$

The state x is an eligible root, a and b are eligible branch presentations, and f is an ineligible failure state:

$$\text{Elig}(x) = \text{Elig}(a) = \text{Elig}(b) = 1, \quad \text{Elig}(f) = 0.$$

Let the witness basis contain a phase witness w_p and an eligibility witness w_e :

$$w_p(x) = \rho, \quad w_p(a) = w_p(b) = \beta, \quad w_p(f) = \emptyset, \quad w_e(z) = \text{Elig}(z).$$

Thus

$$\mathcal{E}(z) = (w_p(z), w_e(z)), \quad \Sigma = \{(\rho, 1), (\beta, 1), (\emptyset, 0)\}.$$

Let $\mathcal{R} = \{\text{id}, \sigma\}$, where

$$\sigma(x) = x, \quad \sigma(a) = b, \quad \sigma(b) = a, \quad \sigma(f) = f.$$

The redescription σ preserves all witness values up to presentation, preserves $\mathcal{E}$ and Elig , and swaps only the branch presentations.

Define two licensed transition schemas $T_a, T_b : S \rightarrow S$ by

$$T_a(x) = a, \quad T_a(a) = a, \quad T_a(b) = a, \quad T_a(f) = f,$$

$$T_b(x) = b, \quad T_b(a) = b, \quad T_b(b) = b, \quad T_b(f) = f.$$

The redescription action on transition labels is

$$\sigma_{\#}(T_a) = T_b, \quad \sigma_{\#}(T_b) = T_a.$$

Then transition covariance holds: $\sigma(T_a z) = T_b(\sigma z)$ and $\sigma(T_b z) = T_a(\sigma z)$ for every $z \in S$. The strict eligible predecessor relation contains $x \prec a$ and $x \prec b$, with x an origin, and is invariant under σ .

The quotient has exactly three interface states:

$$S/\approx = \{[x], [a], [f]\}, \quad [a] = \{a, b\}.$$

The descended signature is

$$\bar{\mathcal{E}}([x]) = (\rho, 1), \quad \bar{\mathcal{E}}([a]) = (\beta, 1), \quad \bar{\mathcal{E}}([f]) = (\emptyset, 0).$$

The quotient transition relation records $[x]\hat{\mathcal{T}}_Q[a]$, $[a]\hat{\mathcal{T}}_Q[a]$, and $[f]\hat{\mathcal{T}}_Q[f]$. The branch outputs are presentation-distinct because $a \not\sim b$, but they are quotient-equivalent because $\sigma(a) = b$.

Suppose an internal admissible selector C chooses one output. If $C(x) = a$, equivariance gives

$$C(x) = C(\sigma x) \sim \sigma(Cx) = \sigma(a) = b,$$

contradicting $a \not\sim b$. The same contradiction follows if $C(x) = b$. The reduced interface therefore records the orbit $\{a, b\}$ as the redescription-neutral branch interface, but it contains no internal fact selecting a rather than b .

The four normal-form exits can also be seen in this toy model. Adding a transported invariant label $\ell(a) \neq \ell(b)$ destroys the symmetry and gives C1. Adding an authorized quotient-level representative convention supplies a canonical section and gives C2, but this is a representative discipline rather than a branch fact internal to the original symmetric system. Adding an outside instruction such as “choose the state printed first on this page” gives C3, because the instruction is not acted on by σ . Sending one branch to f , changing the domain, or resetting eligibility gives C4. Hence the example displays both the obstruction and the exhaustive repair routes, without being a premise of any theorem.

7.4 Contrasting near-failure example

The same finite system also shows that admissibility exclusions are nontrivial. Extend the displayed page by a new datum $p \in \{a, b\}$ saying “preferred printed representative”, and define a selector $C_p(s) = p$. This produces a unique output, but p is not transformed by the swap σ unless it is explicitly added to the state or transition-label signature. If p is not added, the mechanism violates internal data closure and is C3. If p is added but not transformed by $\sigma_\#$, it violates transition-schema covariance. If p is added and transformed, then the original symmetric system has been replaced by one with transported asymmetry, and the case becomes C1. Thus the framework does not declare every convenient selector admissible; it classifies exactly which structural cost the selector pays.

8 Normal Form for Branch-Resolution Attempts

8.1 Resolution mechanisms

Definition 8.1 (Resolution attempt). *A branch-resolution attempt at a symmetric branch point is any procedure that purports to return a unique branch output as the admissible continuation of the same fixed-domain state.*

Definition 8.2 (Internal and external data). *A datum used by a resolution attempt is internal when it is part of the state or fixed transition signature and transforms under every relevant admissible redescription. It is external when it is not acted on by $\mathcal{R}$ but is nevertheless used to decide the branch.*

Lemma 8.3 (External data are parameters). *If a branch-resolution attempt consults a datum not acted on by admissible redescription, then it has injected an external parameter into the fixed-domain evaluation.*

Proof. A datum invisible to $\mathcal{R}$ is fixed across presentation transformations regardless of the transformed state. If it affects the branch verdict, it supplies branch-relevant information not contained in the redescription-neutral state. That is precisely parameter injection. $\square$

8.2 Four primary classes

Definition 8.4 (Resolution classes). *A successful singular branch-resolution attempt belongs to at least one of the following classes:*

- C1. Invariant asymmetry.** *The branches were not genuinely symmetric because a transported admissible-invariant predicate distinguishes them.*
- C2. Canonical section.** *The system adds or already contains a redescription-invariant section choosing canonical representatives of quotient classes or branch orbits.*
- C3. Parameter injection.** *The procedure consults data not transformed by admissible redescrptions.*
- C4. Regime violation or reset.** *The procedure changes the domain, breaks eligibility conservation, violates the envelope, or performs an explicit identity reset rather than a same-domain branch selection.*

Definition 8.5 (Admissibility-priority preorder). *The normal-form classes are ordered by the admissibility-priority preorder*

$$C_4 \succ C_3 \succ C_2 \succ C_1.$$

The ordering is determined by failure depth. Regime violation or reset (C4) removes the procedure from the same fixed domain and therefore absorbs all lower classifications. External parameter injection (C3) supplies the branch authority before any internal representative rule can decide. A canonical section (C2) resolves representation within the regime before invariant asymmetry (C1) is needed as a branch fact. Invariant asymmetry (C1) is the least disruptive same-domain resolution.

Definition 8.6 (Priority-normalized class). *For a resolution attempt instantiating a nonempty set $H \subseteq \{C_1, C_2, C_3, C_4\}$, its primary class is the unique $\succ$ -maximal element of H . Thus a mechanism that both injects a parameter and uses a canonical rule is primarily C3; a mechanism that violates the fixed domain is primarily C4.*

Lemma 8.7 (Hybrid reduction). *Let $H \subseteq \{C_1, C_2, C_3, C_4\}$ be the nonempty set of classes instantiated by a successful resolution attempt. Then H has a unique priority-maximal element. All lower-priority components are either implementation details of that dominant mode or redundant once the dominant mode is present. Therefore no second-order hybrid supplies a fifth same-scope resolution class.*

Proof. The admissibility-priority preorder is linear on the four classes, so every nonempty finite subset has a unique maximum. If C4 is present, the attempt has already left or reset the fixed domain, and lower-priority internal features no longer decide a same-domain branch. If C3 is maximal, the branch-relevant authority is an external parameter; any canonical rule or invariant asymmetry merely processes that imported datum. If C2 is maximal, the decisive ingredient is an authorized section; C1-level features, if present, are redundant for the classification. If only C1 is present, invariant asymmetry is the resolution. Hence hybrids reduce to one dominant locus and do not open a new class. $\square$

Theorem 8.8 (Branch-resolution exhaustion). *Every successful singular branch-resolution attempt at a symmetric fixed-domain branch instantiates at least one of C1–C4. After priority normalization it has exactly one primary class.*

Proof. Let a procedure successfully select a unique branch. If it leaves the fixed domain, changes the envelope, violates eligibility conservation, or resets identity, it is C4. Otherwise it remains inside the same regime. If it consults any branch-relevant datum not acted on by $\mathcal{R}$, Lemma 8.3 makes it C3. Otherwise it uses only internal redescription-acted data. If it selects by imposing a redescription-invariant representative rule or section, it is C2. Otherwise any successful internal

selection must be based on a branch-distinguishing invariant. That is C1. If no such invariant exists, then the procedure is an internal equivariant selector and is impossible by Theorem 7.7. Hence any successful attempt lies in at least one class. Lemma 8.7 then assigns exactly one primary class to any nonempty finite set of applicable classes and rules out a fifth second-order hybrid class. $\square$

Remark 8.9 (Why “exactly one” is normalized). *Raw mechanisms can overlap. For example, a parameterized canonicalization rule is both a parameter injection and a canonical section. The theorem therefore avoids the brittle claim that the raw classes are disjoint. What is exact is the priority-normalized primary class.*

Remark 8.10 (Relative exhaustion). *The exhaustion theorem is relative to fixed-domain admissible evaluation systems satisfying the stated closure, internality, and redescription-stability assumptions. It is not a taxonomy of every possible metaphysical, physical, or computational branch-resolution procedure outside the specified regime.*

9 Non-Duplication of Singular Identity Continuity

9.1 Identity-continuity claims

Definition 9.1 (Singular continuity claim). *Fix an eligible origin $O \in S^+$. A singular continuity claim from O is a redescription-invariant predicate*

$$\text{Cont}_O(Z) \in \{0, 1\}$$

on eligible states, intended to mean that Z is the same singular continuing bearer as O under licensed transport. Redescription invariance requires

$$\text{Cont}_O(Z) = \text{Cont}_{R(O)}(R(Z))$$

for all relevant $R \in \mathcal{R}$.

Definition 9.2 (Duplication attempt). *A duplication attempt from an eligible origin O is a licensed construction producing eligible outputs $X, Y \in S^+$ such that*

$$X \not\sim Y, \quad \text{Cont}_O(X) = 1, \quad \text{Cont}_O(Y) = 1,$$

and no admissible-invariant asymmetry distinguishes X and Y as different continuers. If $X \approx Y$, the quotient treats them as one presentation orbit; if they are treated as two independent singular continuers, an additional invariant distinction or regime change is required.

Lemma 9.3 (Symmetric duplication induces symmetric branching). *A duplication attempt without admissible-invariant asymmetry induces a symmetric branch configuration on the presentation-distinct outputs.*

Proof. The two outputs are presentation-distinct by assumption. Since no admissible-invariant predicate distinguishes them, the local branch orbit is symmetric under the redescription groupoid, or else the obstruction to such symmetry would itself supply an invariant distinction. Thus the two outputs form a symmetric branch configuration at the presentation level. $\square$

Theorem 9.4 (No duplication of singular identity). *In a grounded fixed-domain admissible evaluation system, singular identity continuity from one origin cannot be duplicated across two symmetry-related presentation-distinct eligible outputs without either invariant asymmetry or regime violation.*

Proof. Assume a duplication attempt produces presentation-distinct eligible outputs X, Y without invariant asymmetry. By Lemma 9.3, the situation is a symmetric branch configuration. To maintain singular continuity, the regime must determine which presentation-distinct output is the singular continuer, or else treat two presentation-distinct bearers as one singular identity without quotient collapse.

The first option requires an equivariant selector at a symmetric branch, impossible by Theorem 7.7. The second option either collapses X and Y into the same quotient class, in which case there are not two quotient-level continuers, or transfers one origin's singular standing to multiple presentation-distinct bearers without a fresh invariant distinction. That is a violation of fixed-domain conservation and non-stipulative grounding. Therefore admissible duplication is impossible. Any apparent case must fall into invariant asymmetry, quotient collapse, plural rather than singular continuation, or regime violation/reset. $\square$

Remark 9.5 (Allowed branching versus forbidden duplication). *The theorem does not deny that a process can produce multiple later states. It denies only that two symmetry-related presentation-distinct eligible states can both be independent instances of the same singular continuer of one origin within the same fixed-domain identity regime. Branch plurality is permitted; singular identity duplication is not.*

10 Anchor–Tensor–Skin Decomposition

10.1 Effect tests

The role decomposition is stated as an effect classification on a fixed domain. This keeps the terminology close to standard invariant theory: a role is determined by what changes when the corresponding structure is varied.

Definition 10.1 (Role candidate). *A role candidate is a state component, relation, parameter, or descriptive feature whose variation can be considered while holding fixed the background domain and typing discipline. A variation is admissible only when it is meaningful inside the fixed domain.*

Definition 10.2 (Eligibility effect and transport effect). *Let A be a role candidate.*

- (a) *A has an eligibility effect if varying A while holding other fixed-domain data constant can change Elig.*
- (b) *A has a transport effect if varying A without changing Elig can change the quotient-level transition relation, predecessor, or continuation structure.*
- (c) *A is presentation-only if varying A changes neither Elig nor quotient-level transport/continuation structure.*

Definition 10.3 (Anchor, tensor, skin). *A role candidate has primary role:*

- (i) *anchor if it has an eligibility effect;*
- (ii) *tensor if it has no eligibility effect but has a transport effect;*
- (iii) *skin if it is presentation-only.*

The priority order is anchor before tensor before skin.

Theorem 10.4 (Anchor–tensor–skin decomposition). *Every fixed-domain role candidate has exactly one primary role: anchor, tensor, or skin.*

Proof. Given a role candidate A , either varying A can change eligibility or it cannot. If it can, A is an anchor by definition. If it cannot, then either it can change quotient-level transition relation, predecessor, or continuation structure, or it cannot. In the first subcase it is a tensor; in the second it is skin. The priority convention makes the classification exclusive even if a candidate affects several layers simultaneously. $\square$

Corollary 10.5 (No fourth same-domain admissibility-bearing role). *Within the fixed-domain effect tests above, no fourth primary role exists.*

Proof. The proof of Theorem 10.4 exhausts the possible effects: eligibility effect, transport effect without eligibility effect, or neither. Any alleged fourth role either changes eligibility, changes transport without eligibility, or changes neither. It therefore reduces to anchor, tensor, or skin. $\square$

Corollary 10.6 (Conflation failures). *Conflating the three roles breaks the fixed-domain regime in the following ways:*

- (a) *treating anchors as skin permits presentation variation to change eligibility, violating re-description neutrality;*
- (b) *treating tensors as skin erases quotient-level transport structure, violating transition descent or persistence tracking;*
- (c) *treating skin as anchor overconstrains presentation and blocks harmless redescription, destroying the intended quotient;*
- (d) *treating anchors as tensors or tensors as anchors changes whether a constraint governs eligibility or only continuation, producing a second same-scope eligibility classifier or a hidden repair channel.*

Proof. Each clause follows from the defining effect tests. Anchors are precisely eligibility-affecting constraints, tensors are precisely quotient-transport-affecting constraints below eligibility, and skins are precisely variations killed by the quotient. Interchanging the roles either moves an eligibility effect below the gate, moves transport content into presentation, or promotes presentation into a false eligibility condition. $\square$

11 Rigidity of the Reduced Interface

11.1 Projection layers and morphisms

Definition 11.1 (Reduced projection interface). *For a grounded fixed-domain admissible evaluation system $\mathcal{M}$, its reduced evaluation interface or reduced projection interface is the tuple*

$$\mathcal{I}(\mathcal{M}) = (Q, \bar{\mathcal{E}}, \overline{\text{Elig}}, \hat{\mathcal{T}}_Q, \prec_Q), \quad Q = S/\approx,$$

where $[X] \prec_Q [Y]$ iff representatives can be chosen with $X \prec Y$, and $\hat{\mathcal{T}}_Q$ is the quotient transition relation. This is the formal meaning of “interface” in the paper: a quotient carrier together with descended evaluation signature, descended eligibility, quotient-level transition structure, and quotient-level predecessor structure.

Lemma 11.2 (Quotient predecessor is well-defined). *The relation $\prec_Q$ is well-defined on quotient classes.*

Proof. Suppose $[X] = [X']$, $[Y] = [Y']$, and $X \prec Y$. By definition of $\approx$, choose admissible re-descriptions carrying X to X' and Y to Y' up to $\sim$. Invariance of $\prec$ under admissible redescription gives $X' \prec Y'$ up to $\sim$. Since $\sim$ is identity-preserving, the predecessor relation is unchanged by replacing representatives. Hence the quotient relation is well-defined. $\square$

Definition 11.3 (Admissible interface morphism). *Let*

$$\mathcal{I}_j = (Q_j, \bar{\mathcal{E}}_j, \bar{\text{Elig}}_j, \hat{\mathcal{T}}_{Q,j}, \prec_{Q,j}) \quad (j = 1, 2)$$

be reduced interfaces. An admissible interface morphism

$$(\Phi, \Psi) : \mathcal{I}_1 \rightarrow \mathcal{I}_2$$

consists of a map $\Phi : Q_1 \rightarrow Q_2$ and a signature map $\Psi : \text{im}(\bar{\mathcal{E}}_1) \rightarrow \text{im}(\bar{\mathcal{E}}_2)$ such that for all $q, q' \in Q_1$:

$$(M1) \quad \bar{\mathcal{E}}_2(\Phi(q)) = \Psi(\bar{\mathcal{E}}_1(q));$$

$$(M2) \quad \bar{\text{Elig}}_2(\Phi(q)) = \bar{\text{Elig}}_1(q);$$

$$(M3) \quad q \prec_{Q,1} q' \text{ implies } \Phi(q) \prec_{Q,2} \Phi(q');$$

$$(M4) \quad q \hat{\mathcal{T}}_{Q,1} q' \text{ implies } \Phi(q) \hat{\mathcal{T}}_{Q,2} \Phi(q').$$

It is an admissible conjugacy when Φ and Ψ are bijective and the inverse pair is also an admissible interface morphism. Equivalently, eligibility, evaluation signature, predecessor structure, and quotient transition relation are preserved and reflected.

Definition 11.4 (Category of admissible interfaces). *Let **AdmInt** be the category whose objects are reduced interfaces*

$$\mathcal{I} = (Q, \bar{\mathcal{E}}, \bar{\text{Elig}}, \hat{\mathcal{T}}_Q, \prec_Q)$$

of the kind defined above, and whose arrows are admissible interface morphisms. The identity arrow on $\mathcal{I}$ is $(\text{id}_Q, \text{id}_{\text{im}(\bar{\mathcal{E}})})$. If

$$(\Phi, \Psi) : \mathcal{I}_1 \rightarrow \mathcal{I}_2, \quad (\Phi', \Psi') : \mathcal{I}_2 \rightarrow \mathcal{I}_3,$$

are arrows, their composite is

$$(\Phi', \Psi') \circ (\Phi, \Psi) = (\Phi' \circ \Phi, \Psi' \circ \Psi).$$

*The isomorphisms in **AdmInt** are exactly admissible conjugacies.*

Lemma 11.5 (Admissible interfaces form a category). *The objects and arrows in Definition 11.4 form a category **AdmInt**.*

Proof. The identity pair preserves descended evaluation, eligibility, predecessor, and quotient transition relation immediately. If (Φ, Ψ) and (Φ', Ψ') are admissible interface morphisms, then

$$\bar{\mathcal{E}}_3((\Phi' \circ \Phi)(q)) = \Psi'(\bar{\mathcal{E}}_2(\Phi(q))) = \Psi'(\Psi(\bar{\mathcal{E}}_1(q))),$$

so evaluation preservation is closed under composition. Eligibility preservation, predecessor preservation, and quotient-transition preservation are closed under composition by the same substitution argument. Associativity follows from associativity of ordinary function composition, and identity laws are immediate. If an arrow has a two-sided inverse arrow, both the state-class map and signature map are bijective and preserve and reflect the interface relations; this is precisely admissible conjugacy. $\square$

Definition 11.6 (Representation category over a fixed system). *Fix a grounded system $\mathcal{M}$. Let $\mathbf{Rep}(\mathcal{M})$ be the category whose objects are pairs (J, η) where $J = (Q_J, \bar{\mathcal{E}}_J, \bar{\mathcal{Elig}}_J, \hat{T}_{Q,J}, \prec_{Q,J})$ is an interface-shaped object in $\mathbf{AdmInt}$ and $\eta : S \rightarrow Q_J$ is redescription-invariant, preserves descended eligibility and witness-derived evaluation signatures, and carries quotient-transition consequences as in Proposition 5.10. A morphism*

$$h : (J, \eta) \rightarrow (K, \kappa)$$

is an admissible interface morphism $h : J \rightarrow K$ whose carrier map satisfies $\kappa = h \circ \eta$.

Corollary 11.7 (Categorical form of the minimal quotient interface). *For a grounded fixed-domain admissible evaluation system $\mathcal{M}$, the quotient interface $\mathcal{I}(\mathcal{M})$ equipped with the quotient map $\pi : S \rightarrow S/\approx$ is initial in $\mathbf{Rep}(\mathcal{M})$.*

Proof. For any object (J, η) of $\mathbf{Rep}(\mathcal{M})$, Proposition 5.10 supplies a unique carrier map $\bar{\eta} : S/\approx \rightarrow Q_J$ with $\eta = \bar{\eta} \circ \pi$. The preservation clauses in the definition of $\mathbf{Rep}(\mathcal{M})$ make $\bar{\eta}$ an admissible interface morphism. Uniqueness of the arrow is exactly uniqueness of $\bar{\eta}$ from the surjectivity of π . Therefore $(\mathcal{I}(\mathcal{M}), \pi)$ is initial in the representation category. $\square$

Remark 11.8 (Why this morphism class). *The morphism class is not chosen metaphorically. It is the weakest structure-preserving map that respects every component of the reduced interface: witness-derived signature, binary eligibility, grounded predecessor structure, and quotient transition relation. Dropping any component permits maps that identify states the interface is designed to distinguish.*

Definition 11.9 (Minimal fixed-domain model). *A grounded fixed-domain admissible evaluation system is minimal when its quotient relation is exactly the kernel of its admissibility-relevant invariant content:*

$$X \approx Y$$

if and only if no fixed-domain evaluation signature, anchor test, tensor test, eligibility verdict, predecessor relation, quotient transition relation, or induced stable-transition history distinguishes X from Y . Equivalently, the model contains no extra non-authoritative presentation distinctions above the quotient and no collapsed evaluation, anchor, eligibility, or transport distinctions below it.

Lemma 11.10 (Minimality determines the quotient). *In a minimal model, the admissible quotient $S/\approx$ is uniquely determined by the witness-derived evaluation signatures, anchor tests, tensor tests, eligibility predicate, predecessor relation, and quotient transition relation.*

Proof. Minimality states that two states are quotient-equivalent exactly when all admissibility-relevant invariant tests agree on them. Therefore the equivalence classes are precisely the common fibers of those tests. Common fibers of a fixed family of tests are unique. $\square$

Theorem 11.11 (Local interface rigidity up to admissible conjugacy). *Let $\mathcal{M}_1$ and $\mathcal{M}_2$ be minimal grounded fixed-domain admissible evaluation systems with the same local invariant profile: witness-derived evaluation signatures, anchor tests, tensor tests, eligibility behavior, predecessor structure, and quotient transition relation, all up to relabeling. Then their reduced projection interfaces are isomorphic in the admissible interface-morphism sense, relative to the reduced interface induced by that shared admissibility structure.*

Proof. By Lemma 11.10, the quotient classes in each model are the common fibers of the same admissibility-relevant invariant data, up to relabeling. Define Φ by sending each quotient class of $\mathcal{M}_1$ to the quotient class of $\mathcal{M}_2$ with the corresponding invariant profile, and define Ψ by the corresponding relabeling of signature fibers. The shared invariant data make Φ and Ψ well-defined and bijective. They preserve descended evaluation signatures and eligibility by construction. They preserve and reflect predecessor structure and $\widehat{T}_Q$ because those relations are part of the shared local invariant profile. Hence (Φ, Ψ) and its inverse are admissible interface morphisms, so the reduced projection interfaces are equivalent up to admissible conjugacy. $\square$

Remark 11.12 (What rigidity does and does not say). *Rigidity is not syntactic identity of implementations and it is not a complete classification theorem for all models. The result is local to the reduced projection interface induced by the shared admissibility structure. It says that once evaluation signatures, anchors, and transport invariants are fixed and non-authoritative skin has been quotiented out, there is no second reduced interface performing different admissibility work on the same fixed domain.*

12 Unified Structural Interface Theorem

Theorem 12.1 (Structural interface theorem). *Every non-degenerate fixed-domain evaluative continuity problem satisfying continuity adequacy canonically restricts to a grounded fixed-domain admissible evaluation system and instantiates the kernel of the present constraint architecture. With the AASC full-import closure layer, every such system satisfies the following conclusions:*

- (1) *the framework is a constraint architecture: non-degenerate coherent evaluability forces kernel instantiation rather than selecting one optional model among permissive alternatives;*
- (2) *the admissible evaluation structure is forced by determinate fixed-domain evaluative continuity;*
- (3) *the AMetric boundary blocks pre-fixational selectors, metrics, rankings, probabilities, typicality rules, canonical representatives, and discrete indices unless they are declared invariants of the evaluated domain;*
- (4) *declared boundary traces are the only remaining same-domain role-fixing asymmetry once internal and indexical/provenance routes are neutral;*
- (5) *standing and admissibility collapse to one same-domain gate, so no competing admissibility-bearing classifier remains;*
- (6) *same-scope operators are certified envelopes, and every non-bookkeeping totalization, repair, completion, diagonal, fixed-point, or universal-property detour is inadmissible or a regime change;*
- (7) *governance-bearing variation has no hidden fifth or residual locus beyond boundary typing, the standing partition, witness/trace signature, lawful quotient, admissible continuation, and anchor-tensor-skin role structure;*
- (8) *licensed evaluation factors through a quotient projection;*
- (9) *eligible persistence has at least one non-derivative origin;*

- (10) *symmetric presentation-distinct branching admits no internal equivariant singular representative selector;*
- (11) *every successful singular branch-resolution attempt instantiates one of the normal-form classes $C1$ – $C4$ and has a unique priority-normalized primary class;*
- (12) *singular identity continuity cannot be duplicated across symmetry-related presentation-distinct eligible outputs without invariant asymmetry, quotient collapse, pluralization, or regime violation;*
- (13) *every fixed-domain role candidate decomposes primarily as anchor, tensor, or skin, with no fourth same-domain role;*
- (14) *minimal systems with the same local invariant profiles have reduced projection interfaces isomorphic by admissible interface conjugacy relative to the shared admissibility structure.*

Proof. Item (1) is Corollary 3.12 and Corollary 4.22. Item (2) is Theorem 3.11. Item (3) is Lemma 4.4. Item (4) is Theorem 4.6. Item (5) is Theorem 4.9 and Corollary 4.10. Item (6) is Theorem 4.13 and Corollary 4.14. Item (7) is Theorem 4.16 and Theorem 4.20. Item (8) is Theorem 5.8. Item (9) is Theorem 6.6. Item (10) is Theorem 7.7. Item (11) is Theorem 8.8. Item (12) is Theorem 9.4. Item (13) is Theorem 10.4 together with Corollary 10.5. Item (14) is Theorem 11.11. The full-import closure theorem, Theorem 4.21, states that the boundary, gate, operator, role, trajectory, and hidden-remainder escapes do not add further same-domain cases. $\square$

13 Referee-Facing Audit

13.1 Dependency map

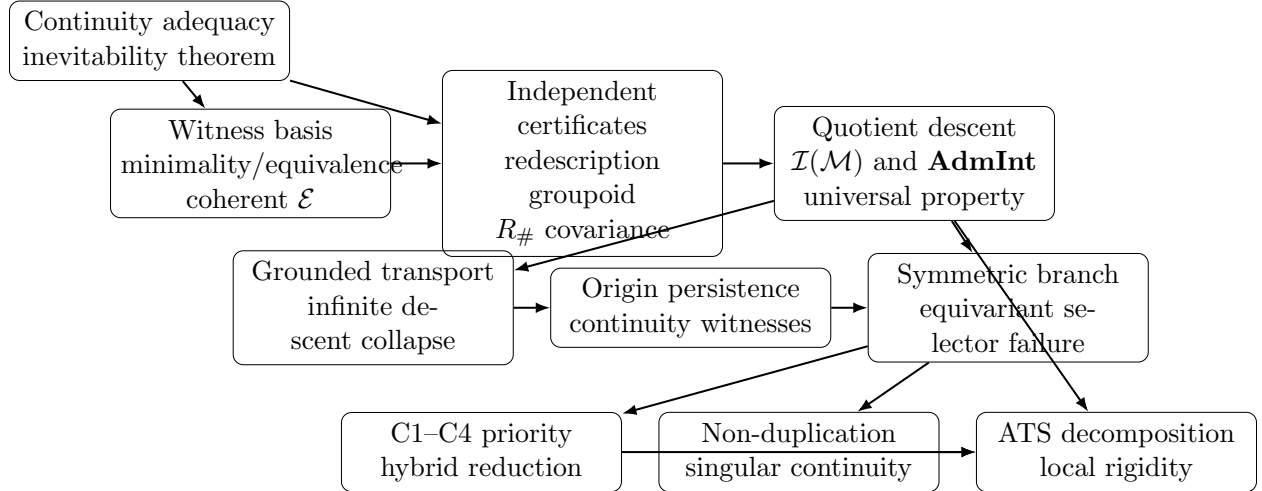
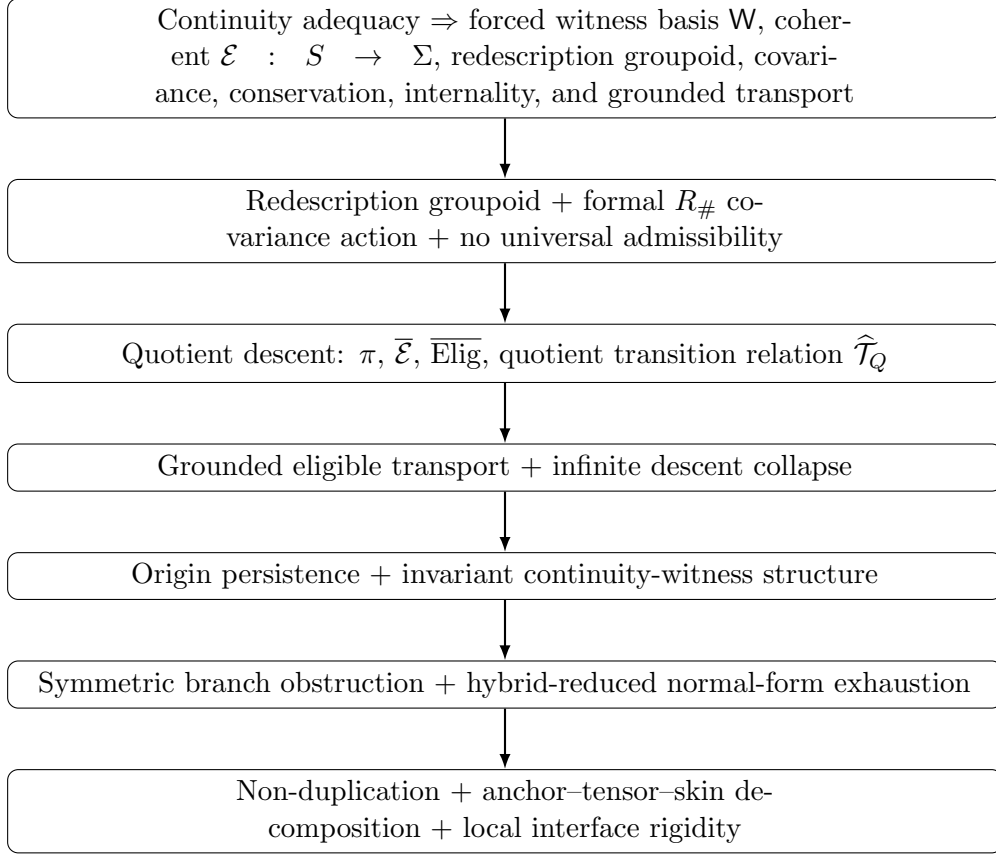


Figure 1: Visual-first dependency DAG. The proof spine begins with witness-constrained evaluation and admissibility certificates, descends through quotient formation, and then branches into transport grounding, symmetry obstruction, normal-form exhaustion, non-duplication, role decomposition, and rigidity.



Result	Load-bearing assumptions	Main failure if removed
Quotient descent	Witness-coherent evaluation; re-description groupoid; transition covariance	Presentation dependence; no well-defined quotient transition structure
Witness-basis equivalence	Consequence-equivalent witness bases; redundancy invariance; witness coherence	Quotient interface depends on arbitrary extractor enumeration rather than invariant consequence kernel
Origin necessity	Nonempty eligible domain; infinite descent collapse; well-founded strict predecessor relation	Infinite regress or arbitrary standing labels
Selector impossibility	Symmetric branch; equivariance; presentation-distinct outputs	Hidden asymmetry or parameterized choice
Resolution exhaustion	Internal data closure; selector impossibility; fail-closed fixed domain; hybrid reduction	Unclassified external choice, domain drift, or alleged fifth hybrid
Non-duplication	Singular continuity; branch symmetry; conservation	Plural branches mislabeled as one singular identity
Anchor–tensor–skin	Fixed-domain effect tests	Fourth-role claims become equivocal
Local rigidity up to admissible conjugacy	Minimality; shared evaluation and invariant profiles; quotient descent; interface morphisms; AdmInt category	Redundant skin or collapsed invariant structure
Categorical universality	Presentation-independent representation; quotient universal property; admissible interface morphisms	Factorization exists as notation only, not as an interface-level universal property

13.2 Standard objections and exact burdens

Objection	What would have to be shown
“The witness basis is arbitrary.”	Two consequence-equivalent witness bases that generate non-conjugate reduced interfaces, or a nonredundant witness that changes no fixed-domain admissibility-relevant consequence but cannot be deleted without changing the interface.
“The universal property has no category.”	A presentation-independent admissibility-preserving representation satisfying the stated hypotheses whose factorization is not an admissible interface morphism in AdmInt .
“The result is definitional.”	A system satisfying witness minimality or redundancy reduction, coherence, fixed-domain invariance, formal transition covariance, conservation, internality, independent admissibility certification, no universal admissibility, and grounded transport while failing one of the seven conclusions.
“A symmetric branch can still be selected.”	A redescription-equivariant selector at a transitive two-branch orbit with presentation-distinct outputs.
“Persistence can be derivative all the way down.”	A nonempty well-founded eligible predecessor relation with no minimal eligible element, or a non-well-founded relation that still supplies a non-stipulative terminating transport witness.
“There is a fourth role beyond anchor, tensor, and skin.”	A fixed-domain feature that neither affects eligibility, nor affects quotient transport below eligibility, nor is presentation-only.
“Duplication is possible.”	A pair of symmetry-related presentation-distinct eligible outputs both bearing one singular identity-continuity relation as independent continuers without selecting, quotient-collapsing, pluralizing, injecting a parameter, or violating the regime.
“Rigidity overstates implementation identity.”	The theorem does not claim implementation identity or complete model classification; a counterexample must preserve all invariant profiles while producing a reduced quotient interface not equivalent up to admissible conjugacy.

14 Relation to Existing Structural Frameworks

The construction overlaps with several familiar structural frameworks but is not identical to any of them. Unlike ordinary equivalence frameworks, admissibility here is constrained jointly by evaluative coherence, transport covariance, witness stability, AMetric boundary discipline, standing-admissibility gate identity, operator-envelope certification, fail-closed eligibility, independent witness certification, grounded predecessor structure, and fixed-domain internality. It differs from ordinary automata theory because the central object is not merely a transition system; it is a transition system with witness-coherent evaluation, eligibility conservation, redescription covariance, and a quotient interface forced by admissible presentation change. It differs from bisimulation because the quotient relation is not chosen only to match external behavior; it is generated by identity-preserving redescription and then constrained by witness extraction, eligibility, predecessor, and transition structure. It differs from gauge redundancy in physics because no physical gauge group is assumed; gauge-like removal of presentation surplus appears here as one instance of a more

general fixed-domain quotient requirement. It differs from modal or structuralist metaphysics because the claims are conditional theorems about explicitly specified systems rather than claims about what objects ultimately are. Finally, it differs from category-theoretic equivalence in general because the admissible morphisms preserve a particular reduced-interface profile: witness-derived signatures, binary eligibility, grounded predecessor structure, and quotient transition relation.

This positioning matters for the interpretation of the results. The paper does not claim that quotienting, bisimulation, or structural equivalence is new. The contribution is the combined closure: once witness-coherent evaluation, redescription covariance, fail-closed eligibility, internality, grounded transport, and admissible morphisms are imposed together, persistence, equivariant selector failure, normal-form exhaustion, non-duplication, role decomposition, and local interface rigidity follow as one constrained package. In that sense the manuscript is not a permissive descriptive taxonomy. It is a constraint architecture: rival structures remain admissible only if they instantiate the same kernel up to harmless equivalence; otherwise they fail one of the fixed-domain coherence conditions or change the problem.

15 Discussion

The inevitability result changes the logical status of the framework. The paper is not merely the study of systems that happen to satisfy admissibility, coherence, covariance, internality, and grounding. It shows that once a fixed-domain problem demands determinate evaluative continuity under presentation neutrality and finite witness extraction, those structures are the minimal stable form of the problem. More precisely, the paper should be read as a constraint architecture: non-degenerate coherent evaluability requires instantiation of the kernel. A different choice is possible only by weakening one of the preservation demands, accepting a classified failure mode, or explicitly changing regimes.

The AASC full-import closure layer narrows the remaining alternatives further. It shows that the boundary cannot perform selection by a hidden metric or representative convention, that standing and admissibility cannot be separated into rival gates, that same-scope operators cannot totalize or repair the regime from inside, that diagonal and fixed-point detours add no new admissible layer, and that no hidden governance-bearing locus remains after boundary, quotient, trajectory, and role loci are fixed.

This is why the manuscript eliminates alternatives rather than cataloguing them. The eliminations are not imposed by preference; they are the effect of constraint propagation. Once the fixed-domain kernel is required to support coherent evaluability, a purported same-domain alternative must either realize the same kernel under a different presentation or exhibit a determinate failure mode: hidden parameter, non-neutral presentation, unsupported boundary choice, competing gate, illicit operator envelope, post hoc repair, role conflation, or regime change.

The formal content can be summarized without mentalistic vocabulary. A stable fixed-domain evaluation system with a witness-coherent evaluation functional, identity-preserving presentation change, and redescription-covariant transition schemas admits a quotient transition interface. A non-vacuous well-founded positive region has origins. Symmetric branches have no equivariant singular selector. Successful branch resolution must add asymmetry, a canonical section, an external parameter, or leave the fixed domain. Singular continuity cannot be duplicated across symmetry-related presentation-distinct outputs without invariant asymmetry, quotient collapse, pluralization, or regime violation. Fixed-domain roles reduce by effect to eligibility constraints, transport content, and presentation surplus. Minimal reduced interfaces are unique up to admissible conjugacy under shared invariant data.

The canonicity claims are correspondingly local. The framework does not require a unique literal witness list, a unique syntactic implementation, or a unique presentation of the signature codomain. It requires that admissibility-relevant consequences be fixed. Once consequence-equivalent witness bases, redescription-covariant transition schemas, grounded predecessor structure, and admissible morphisms are fixed, the reduced quotient interface is invariant up to admissible conjugacy.

The reason these results may be described as mind-like is purely structural: they reproduce an interface role profile associated with persistence, perspective, self-location, non-duplication, and substrate-independent presentation. The framework classifies invariant continuity structures within admissible evaluation systems and does not assume any independent metaphysical theory of persons, consciousness, or ontology. The argument does not claim that these roles are sufficient for conscious experience. It claims only that any system satisfying the stated invariance and conservation conditions already has this constrained interface geometry.

A Notation Crosswalk

Earlier phrase	Mainstream-facing phrase used here	Formal object
Standing	Eligibility / standing-positive status	$\text{Elig} : S \rightarrow \{0, 1\}$
Admissible redescription	Identity-preserving presentation transformation	$R \in \mathcal{R}$
AER*	Fixed-domain admissible evaluation system	$\mathcal{M} = (S, \sim, \mathcal{W}, \mathcal{E}, \mathcal{R}, \mathcal{T}, \text{Elig})$
Evaluation content	Witness-coherent evaluation signature functional	$\mathcal{W}, \mathcal{E} : S \rightarrow \Sigma$
Projection / mediator	Quotient projection	$\pi : S \rightarrow S/\approx$
Persistence origin	Minimal element of well-founded eligible predecessor relation	$O \in S^+$ with no $X \prec O$
Self-location selector	Equivariant representative choice function	$C(RX) \sim R(CX)$
Escape classes	Normal-form classes for branch resolution	C1–C4
Anchor	Eligibility-affecting invariant constraint	effect on Elig
Tensor	Quotient-level transport content	effect on $\hat{\mathcal{T}}_Q$ or $\prec_Q$
Skin	Presentation-only quotient-killed variation	no eligibility or transport effect
Transition covariance	Transition-schema covariance under re-description	$R(TX) \sim (R\#T)(RX)$
Interface equivalence	Admissible interface conjugacy	morphisms preserving $\bar{\mathcal{E}}, \bar{\text{Elig}}, \hat{\mathcal{T}}_Q, \prec_Q$
Interface category	Category of admissible interfaces	AdmInt
Witness-basis canonicity	Consequence-equivalent bases give the same interface	Theorem 2.6

B Tightened Proof Seams

1. **Evaluation functional.** Evaluation is represented by a witness-extraction basis $\mathcal{W}$ and coherent signature map $\mathcal{E} : S \rightarrow \Sigma$; both $\mathcal{E}$ and Elig descend to the quotient.
2. **Evaluation coherence.** The signature space Σ is the image of fixed-domain witness consequences, not an arbitrary codomain chosen to force a result.

3. **Witness minimality.** Redundant witnesses may be deleted without changing the reduced interface; nonredundant witnesses mark real regime content rather than hidden semantic padding.
4. **Admissibility nontriviality.** Maps that change eligibility, change witness consequences, use external parameters, lack independent witness certificates, or fail covariance are excluded; admissibility is not the class of all self-maps.
5. **No universal admissibility.** Non-degeneracy and fail-closed witness coherence rule out treating every well-typed self-map as admissible.
6. **Redescription equivalence.** The inverse and composition requirements on $\mathcal{R}$ are explicit, so $\approx$ is an ordinary orbit equivalence rather than a relation whose symmetry is assumed informally.
7. **Transition covariance.** Branch transitions may be permuted by admissible redescription through the formal $R_{\#}$ action; stable individual transitions are a special case that descend to maps.
8. **Transport grounding.** The grounding theorem makes explicit why non-well-founded eligible transport yields audit failure, stipulative continuity, or absence of an invariant continuity witness.
9. **Persistence rank.** The rank used for persistence is the standard ordinal rank of a well-founded relation. The proof no longer assumes that an arbitrary predecessor of Y lies on a shortest chain to Y .
10. **Identity transition.** The strict predecessor relation excludes the identity transition, avoiding the collapse in which every eligible state becomes derivative from itself.
11. **Symmetric choice.** The self-location result is the standard nonexistence of an equivariant selector on a transitive branch orbit.
12. **Exhaustion.** Raw escape classes may overlap, so exactness is stated only after priority normalization.
13. **Hybrid reduction.** Mixed resolution attempts reduce to a unique dominant class under the formal $C4 > C3 > C2 > C1$ admissibility-priority preorder, closing second-order hybrid evasions.
14. **Role decomposition.** Anchor, tensor, and skin are determined by fixed-domain effect tests, making the trichotomy checkable rather than metaphorical.
15. **Witness-basis equivalence.** Consequence-equivalent witness bases induce equivalent reduced interfaces, so witness choice is canonical up to invariant consequence structure rather than canonical as a syntactic list.
16. **Interface morphisms.** Admissible conjugacy is defined as an isomorphism preserving descended evaluation, eligibility, quotient transition relation, and predecessor structure.
17. **Interface category.** Reduced interfaces and admissible interface morphisms form the category **AdmInt**, and the quotient interface is initial in the corresponding representation category over a fixed system.

18. **Universal property.** Any presentation-independent admissibility-preserving representation factors uniquely through the quotient interface.
19. **Rigidity.** Minimality is defined as equality between quotient equivalence and the kernel of all admissibility-relevant invariant tests, so admissible conjugacy of reduced interfaces follows from shared invariant profiles.
20. **AMetric boundary.** Boundary-side selection, metrics, rankings, probability weights, typicality rules, and canonical representatives have no admissibility-bearing authority unless declared as transported invariants.
21. **Boundary trace fixation.** When internal and provenance/indexical asymmetries are neutral, only a singleton stable declared trace can fix a role; post-selection traces are repair or scope reset.
22. **Gate identity.** Standing and admissibility are not rival predicates on the same fixed domain; every separation route collapses.
23. **Operator exhaustion.** Completion, totalization, forcing, saturation, diagonalization, fixed-point, universal-property, and foundation-swap moves are either bookkeeping-equivalent envelopes or inadmissible same-domain operators.
24. **No-hidden-remainder.** Any governance-bearing difference changes boundary typing, the standing partition, witness/trace content, quotient structure, continuation structure, or role structure; otherwise it is skin/bookkeeping or scope change.

C AASC Import Ledger for v8

The v8 closure pass used the AASC bundle as a theorem-tool library. The imports are integrated as self-contained statements in Section 4; this ledger records the role each AASC strand plays in the present manuscript.

AASC strand	Imported hardening role in this manuscript
AMetric boundary	Blocks pre-boundary selectors, metrics, rankings, probabilities, typicality rules, canonical representatives, boundary trajectories, and backward transport of interior operators.
Boundary-trace fixation	Supplies the declared-trace normal form for role fixation after internal and indexical/provenance asymmetries are exhausted.
Fixed-domain exhaustion	Supplies no-hidden-remainder: every governance-bearing variation changes boundary typing, standing partition, witness/trace content, lawful quotient, continuation, or role structure.
Standing-admissibility identity closure	Collapses same-domain separation routes between standing and admissibility and rules out competing gates.
Same-scope operator closure	Rules out totalization, completion, repair, forcing, saturation, diagonalization, fixed-point, universal-property, and foundation-swap operator escapes.
Unus solus / unique admissible interior	Strengthens the no-generators, no-carriers, no-repairs, no-scope-transport, no-competing-gates, and uniqueness-of-interior readings.
Anchor–tensor–skin decomposition	Strengthens the effect-test trichotomy into no-fourth-role fixed-domain closure.
Coherent construction trajectory theory	Supplies retained-trajectory closure, no-deferred coherence, failure-witness persistence, and the reading of evaluative continuity as the retained-history locus.
Gödel/non-instantiability closure	Blocks same-domain self-reference, diagonal, encoding-multiplicity, and meta-foundational rescue routes once the fixed admissibility substrate is in force.